

Efficient Algorithms for Infinite-Dimensional Spectral Computations



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Abstract

In recent work, Matthew Colbrook has developed a new approach for computing the spectra and pseudospectra of infinite-dimensional operators. Under certain conditions on the operator, this approach yields convergent approximations that do not suffer from the common pitfalls of the standard discretize-then-solve approach.

In this dissertation, we develop an adaptive finite element realization of Colbrook's approach for computing the pseudospectra of infinite-dimensional operators. This algorithm iterates over test points in the complex plane, solves two generalized eigenproblems at each test point, and uses the smallest computed eigenvalues to approximate the pseudospectra of the underlying operator. Since this algorithm requires repeated eigensolves over potentially many test points, its implementation can be computationally expensive.

In an effort to improve both the accuracy and efficiency of Colbrook's approach, we employ adaptivity on two levels. We use the dual-weighted residual (DWR) method to drive an adaptive mesh refinement technique for the auxiliary eigenproblems at each test point. To reduce the number of eigensolves required for accurate pseudospectral computations, we develop a method for adaptively selecting test points in the complex plane. Our main algorithm uses these two levels of adaptivity to focus the computational effort of Colbrook's approach near the pseudospectral contour of interest. We demonstrate the success of this coupling by applying it to a one-dimensional Dirac-type operator and to the two-dimensional Maxwell eigenproblem.

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Chapter 1

Introduction

1.1 Motivations and Project Aims

It is hard to overstate the importance of spectra in applied mathematics. Trefethen notes that at least twenty Nobel Prizes in Physics have been awarded for phenomena associated with eigenvalues [29]. Spectra play a major role in numerous fields, such as acoustics, chemical analysis, dynamical systems, fluid mechanics, quantum mechanics, and partial differential equations¹. A particularly interesting example can be found in spectroscopy. As displayed in Figure 1.1, the black ‘Fraunhofer lines’ that appear in spectroscopic images of the sun’s light correspond to the differences of eigenvalues of the Schrödinger operator for atoms such as H, Fe, Ca, Na, and Mg [30]. Spectroscopic measurements such as these allowed for the measurement of redshifts of distant galaxies, leading to the discovery of the expanding universe [30].

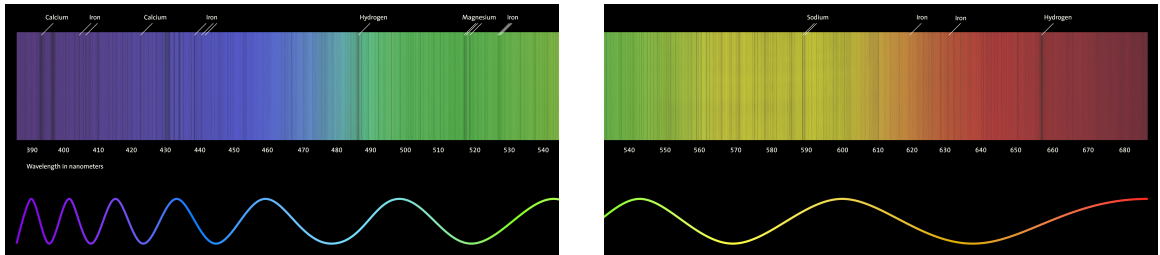


Figure 1.1: Spectroscopic images of the sun’s light, courtesy of the Griffith Observatory [15, 16].

Although the importance of spectra is beyond question, their reliable computation can be challenging [26]. In finite dimensions, we compute the spectrum of a matrix, which is equivalent to the set of its eigenvalues. In particular, if $B \in \mathbb{C}^{n \times n}$ then its spectrum is the following set of points:

$$\text{Sp}(B) = \{z \in \mathbb{C} : \det(B - zI) = 0\} = \{z \in \mathbb{C} : (B - zI)^{-1} \text{ does not exist}\}, \quad (1.1)$$

where I denotes the identity matrix. In infinite dimensions, we consider a linear operator A on a Hilbert space $\mathcal{H}$, whose spectrum is defined as

$$\text{Sp}(A) = \{z \in \mathbb{C} : (A - zI)^{-1} \text{ does not exist as a bounded operator on } \mathcal{H}\}, \quad (1.2)$$

¹For an overview of the history and applications of eigenvalue problems, which are a subset of spectral problems, see the introduction of [30] and the references therein.

where I denotes the identity operator. The spectrum of a linear operator includes the set of eigenvalues of A , but is generally much richer.

Patrick: You should define eigenvalues here, give the formula. Not everyone understands sharply the distinction between spectra and eigenvalues.

This follows from the fact that in infinite dimensions there are more ways for $A - zI$ to fail to have a bounded inverse [9].

Patrick: The Big Book Principle: when citing a specific result in a big book, don't ever just cite the whole book. Tell your readers where to find what you're talking about, like in Chapter 1, or Theorem 2.5, or whatever.

As a result, new phenomena are present in the spectra of linear operators that make them much harder to reliably compute. To see this, consider the right-shift operator,

$$A(x_1, x_2, \dots) = (0, x_1, x_2, \dots), \quad (1.3)$$

that acts on square-summable sequences². This operator's spectrum is the unit disc, but it has no eigenvalues³.

The standard approach for computing the spectrum of an infinite-dimensional operator is often referred to as the *discretize-then-solve* approach [9].

Patrick: Big Book Principle

This approach consists of discretizing the operator with a finite-dimensional matrix, computing the spectrum of the discretized operator, and hoping that the spectrum of the discretized operator approaches the spectrum of the infinite-dimensional operator as the size of the discretization increases. Despite being a successful approach for many problems, it is fraught with difficulties. If the problem is not discretized very carefully, *spectral pollution* can occur, where the discretized operator has eigenvalues that converge to something not in the spectrum of the infinite-dimensional operator.

Patrick: Need to be more precise here: it is a sequence of operators that has a sequence of eigenvalues that converges

The approach can also suffer from *spectral invisibility*, where discretizing the operator causes us to miss parts of the spectrum, even as the size of the discretization increases. Even if the discretization manages to avoid both spectral pollution and spectral invisibility, there remains a far more catastrophic limitation of the discretize-then-solve approach. In particular, since the spectra of infinite-dimensional operators

²The set of square-summable sequences is $\ell^2(\mathbb{N}) = \{z = \{x_n\}_{n \in \mathbb{N}} : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$.

³For more information on this example, see Appendix A.

may include more than just eigenvalues of finite multiplicity, there may be parts of the spectrum that we can never capture with the finite set of eigenvalues that results from discretization [9].

Patrick: Please write in British English, e.g. 'discretise' rather than 'discretize'. You are a loyal subject of the Crown, after all? AI can do this automatically.

These drawbacks of the discretize-then-solve approach motivate the development of alternative methods for computing the spectra of infinite-dimensional operators. This challenge has been taken up in the recent work of Matthew Colbrook [9].

Patrick: E.g. this big book citation is fine.

His work is based on the idea of deferring discretization to the last opportunity: he devises algorithms that compute spectra at the infinite-dimensional level, then discretizes the subproblems. [26].

Colbrook's algorithm for computing spectra begins with his algorithm for computing pseudospectra, so we consider this first. This algorithm considers a sequence of complex numbers z , and computes bounds on the distance from z to $\text{Sp}(A)$. These bounds are obtained from the smallest spectral values of the *folded operators* $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$, where $(A - zI)^*$ is the adjoint operator of $A - zI$.

Patrick: A is still fixed in notation to be the right-shift operator. You need to redefine A to be an endomorphism on a Hilbert space.

Discretization enters Colbrook's algorithm at this stage, where the smallest spectral values of the folded operators are approximated by the smallest eigenvalues of their finite-dimensional discretizations. The benefit of discretizing the folded operators as opposed to the original operator is that, when A is closed and densely defined, the folded operators are non-negative and self-adjoint. The smallest spectral value of a non-negative self-adjoint operator is typically much easier to compute than other parts of its spectrum, and is less prone to the pitfalls of the discretize-then-solve approach [9].

Under certain conditions on the operator, Colbrook's algorithm yields a convergent approximation of the spectrum that captures the continuous components of the spectrum and does not suffer from spectral pollution or spectral invisibility. This success, however, comes at a significant computational cost because it requires solving eigenproblems at potentially many points in the complex plane. Furthermore, as we shall see, coarse discretizations of the folded operators can produce poor approximations of the bound on the distance to the spectrum. As a result, further refinement, and thus

more expensive computations, may be needed to obtain meaningful approximations.

In this dissertation, we explore methods for increasing the efficiency of Colbrook’s algorithm for computing the spectra of infinite-dimensional operators.

Patrick: Really we compute pseudospectra?

We first aim to exploit the structure of the auxiliary eigenproblems in Colbrook’s algorithm in an effort to solve them efficiently. To do so, we apply the *dual-weighted residual* (DWR) method to estimate the error in the quantities used to compute the spectrum.

Patrick: Add something like: This employs the adjoint or dual eigenproblem to estimate the discretisation error in the computation of goal functionals; in this case the goal functional is the lowest eigenvalue of the folded operator ...and then cite Giles & Süli, Giles & Pierce, Becker & Rannacher ...

These estimates are used to both improve the computed approximations and to drive adaptive refinement of the auxiliary eigenproblems.

Patrick: Say: We refer to this adaptivity, changing the mesh employed at every point tested in the complex plane, as “inner-adaptivity”.

In an effort to reduce the number of eigenproblems that must be solved, we also develop a method for adaptively selecting grid points in the complex plane. This “outer-adaptivity” balances the DWR error estimate with the desired spatial resolution of a pseudospectral computation.

To investigate the effectiveness of our methodology, we apply it to numerical examples implemented using the finite element method. In doing so, we aim to show that pairing Colbrook’s algorithm with an adaptive methodology enables more efficient and reliable computations of the pseudospectra of infinite-dimensional operators.

1.2 Main Contributions

The main contributions of this work can be summarized as follows:

- We specialize Colbrook’s algorithm for computing pseudospectra to the finite element setting.
- We develop an adaptive mesh refinement technique that applies the DWR method for symmetric eigenproblems to the auxiliary eigenproblems that arise in Colbrook’s algorithm. To do so, we extend the current DWR eigenvalue error estimate to eigenvalues of multiplicity greater than one. We also use the DWR

eigenvalue error estimate to derive an error estimate for Colbrook’s approximate bounds on the distance to the spectrum.

Patrick: What does ‘current’ mean? Make it clear you have improved the formulae published in the literature.

- We develop a methodology for adaptively selecting test points for Colbrook’s algorithm

Patrick: to compute a specified pseudospectral contour? It’s stated later but seems more natural here

. To do so, we triangulate a region of the complex plane and use our DWR error estimates to derive criteria for guaranteeing whether a given triangle lies inside or outside of the desired pseudospectral contour, or whether it should be further refined.

- We couple an adaptive mesh refinement technique for the auxiliary eigenproblems with our adaptive selection of test points. This coupling focuses the computational effort of Colbrook’s algorithm near the pseudospectral contour of interest.
- We apply the resulting methodology to a one-dimensional Dirac-type operator and to the Maxwell eigenproblem.

Patrick: I for one wouldn’t know what a ‘Dirac’ operator is, maybe explain

1.3 Code Availability

All code is implemented using the complex version of Firedrake [17]. Our implementation of an adaptive mesh refinement technique for Colbrook’s algorithm builds upon an existing Firedrake branch that provides a general implementation of DWR-based adaptivity for nonlinear variational problems [13].

Patrick: This citation should credit Pablo Brubeck

To this branch, we updated the DWR-based adaptive eigensolver developed by Joseph Flood for his MMSC thesis last year [14]. We then extended both implementations so that they can be applied to the auxiliary eigenproblems in Colbrook’s algorithm.

The code used to produce each numerical example presented below is available at [LINK](#). The filenames indicate the relevant figure numbers.

1.4 Structure of the Dissertation

The remainder of the dissertation is structured as follows. In Chapter 2, we present all relevant background material for our methodology. We begin with an overview of the spectra and pseudospectra of infinite-dimensional operators. We then introduce the standard discretize-then-solve approach and discuss its common pitfalls. As an example, we demonstrate spectral pollution when a naïve finite element discretization is applied to the Maxwell eigenproblem. This motivates Colbrook’s algorithm for computing pseudospectra, which we first describe in its abstract formulation and then in the finite element setting. By applying the finite element realization of Colbrook’s algorithm to the Maxwell eigenproblem, we show that, although the algorithm avoids the pitfalls the discretize-then-solve approach, its accuracy critically depends on the quality of the underlying discretization. As a result, we introduce the Dual-Weighted Residual (DWR) method

Patrick: Stop defining the acronym over and over, you’ve defined it already, just use it

and explore its application to our finite element realization of Colbrook’s algorithm.

In Chapter 3, we develop our adaptive methodology. We first construct an adaptive mesh refinement technique for the auxiliary eigenproblems arising in Colbrook’s algorithm. We then develop an approach for adaptively selecting test points within it. Chapter 3 ends with our main new algorithm, which couples these two levels of adaptivity.

In Chapter 4, we apply our adaptive algorithm to a one-dimensional Dirac-type operator and to the Maxwell eigenproblem. In Chapter 5, we summarize our work and discuss some potential directions for future research.

Chapter 2

Background

In this chapter, we present all necessary background material. We begin with an overview of the relevant definitions and results from spectral theory. Then, we describe the common pitfalls of the discretize-then-solve approach. To see one of these pitfalls in action, we consider the famous Maxwell eigenproblem, which exhibits spurious eigenvalues when discretized with standard linear Lagrange finite elements.

Having explored the limitations of the discretize-then-solve approach, we introduce Matthew Colbrook's algorithm for computing the spectra of infinite-dimensional operators. When applied to the Maxwell eigenproblem, this approach avoids spurious eigenvalues, even when discretized with linear Lagrange finite elements. However, when the discretization is coarse, Colbrook's approximate bounds on the distance to the spectrum may be poor. In an effort to quantify the error in these approximations, and to improve them without extensive uniform refinement, we explore the Dual-Weighted Residual (DWR) method for eigenproblems and investigate its application to Colbrook's algorithm.

2.1 An Introduction to Spectral Theory

In this section, we provide an overview of the spectral theory required for this project. In the first subsection, we recall several standard results from functional analysis and introduce the basic concepts needed to define the spectrum of a linear operator. In the second subsection, we extend this discussion to the ϵ -pseudospectrum of an operator.

2.1.1 The Spectra of Linear Operators

Let $\mathcal{H}$ be a Hilbert space with inner product $(\cdot, \cdot)_{\mathcal{H}}$ and norm $\|\cdot\|_{\mathcal{H}}$. We consider a linear operator $A : D(A) \rightarrow \mathcal{H}$ defined on a subspace $D(A) \subset \mathcal{H}$. We use $D(A)$ and $R(A)$ to denote the domain and range of A , respectively. Furthermore, we use $N(A)$ to denote the null space, or kernel, of A .

We say that A is *bounded* on $\mathcal{H}$, denoted $A \in \mathcal{B}(\mathcal{H})$, if there exists a constant $c > 0$ such that $\|Ax\|_{\mathcal{H}} \leq c\|x\|_{\mathcal{H}}$ for all $x \in D(A)$. When a bounded linear operator's range lies on the real line or in the complex plane, it is called a *bounded linear functional* [22]. Furthermore, we say that A is *boundedly invertible* if there exists a bounded operator $B : \mathcal{H} \rightarrow D(A)$ such that $AB = I$ and $BA = I$, where I denotes the identity operator. In this case, B is called the *inverse* of A [9].

We say that A is *closed* if and only if it has the following property: whenever

$\{x_n\}_{n \in \mathbb{N}}$ is a sequence in $D(A)$ such that $x_n \rightarrow x$ and $Ax_n \rightarrow y$, then $x \in D(A)$ and $Ax = y$ [22]. We note that when A is closed, being boundedly invertible is equivalent to $N(A) = \{0\}$ and $R(A) = \mathcal{H}$.¹

To introduce the *Hilbert adjoint* of A , we assume that A is *densely defined*, i.e., that $D(A)$ is dense in $\mathcal{H}$. Furthermore, we define

$$D(A^*) = \{w \in \mathcal{H} : v \mapsto (Av, w)_H \text{ is a bounded linear functional on } D(A)\}. \quad (2.1)$$

Consider $w \in D(A^*)$. Since $D(A)$ is dense in $\mathcal{H}$, there exists a unique $u \in \mathcal{H}$ such that

$$(Av, w)_H = (v, u)_H, \quad \forall v \in D(A). \quad (2.2)$$

The Hilbert adjoint of A is defined as the linear operator A^* that satisfies $u = A^*w$. When $A^* = A$, the linear operator A is said to be *self-adjoint*. Furthermore, when A is closed and densely defined, A^*A and AA^* are self-adjoint. In this case, if $AA^* = A^*A$, then A is called *normal*.

In this setting, the spectrum of A , as stated in (1.2), can be equivalently formulated as the following set:

$$\text{Sp}(A) = \{z \in \mathbb{C} : A - zI \text{ is not boundedly invertible}\}. \quad (2.3)$$

If A is not a closed linear operator, then $\text{Sp}(A) = \mathbb{C}$. Thus, going forward, we assume that A is a closed linear operator. In this case, $\text{Sp}(A)$ is a closed subset of the complex plane [9].

As is the case in finite dimensions, $z \in \mathbb{C}$ is an *eigenvalue* of A if $A - zI$ is not injective, i.e., if $N(A - zI) \neq \{0\}$. We use $\Gamma(A)$ to denote the set of the eigenvalues of A . For each $z \in \Gamma(A)$, the corresponding nullspace of $A - zI$ is the *eigenspace* of A corresponding to z . Notice that $\Gamma(A) \subset \text{Sp}(A)$ but that there may exist $z \in \text{Sp}(A) \setminus \Gamma(A)$. For example, $A - zI$ may fail to be boundedly invertible because $R(A - zI)$ is not all of $\mathcal{H}$, even when z is not an eigenvalue of A .

If the linear operator A is closed and densely defined, the spectrum of A can be related to the spectrum of its Hilbert adjoint A^* . In this case, $\text{Sp}(A^*) = \{\bar{z} : z \in \text{Sp}(A)\}$, where $\bar{z}$ denotes the complex conjugate of $z \in \mathbb{C}$. Moreover, if $z \notin \text{Sp}(A)$ then $(A - zI)^{-1}$ exists and $[(A - zI)^{-1}]^* = (A^* - \bar{z}I)^{-1}$.

One of the most important tools for the analysis and computation of spectra is the resolvent operator:

$$(A, z) \mapsto (A - zI)^{-1} \quad z \notin \text{Sp}(A). \quad (2.4)$$

¹When $N(A) = \{0\}$ and $R(A) = \mathcal{H}$, A^{-1} is defined. Since A is closed so is A^{-1} . Then, the Closed Graph Theorem tells us that A^{-1} is bounded [22].

The norm of the resolvent provides information about the distance from $z \notin \text{Sp}(A)$ to the spectrum of A . In particular,

$$\|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} \leq \text{dist}(z, \text{Sp}(A)), \quad (2.5)$$

where

$$\text{dist}(z, \text{Sp}(A)) = \inf_{w \in \text{Sp}(A)} |z - w|. \quad (2.6)$$

If the operator A is normal, we have equality:

$$\|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} = \text{dist}(z, \text{Sp}(A)). \quad (2.7)$$

Hence, when A is a normal operator, the norm of the resolvent can be used to characterize the spectrum of A .

2.1.2 The Pseudospectra of Linear Operators

To ask whether a $z \in \mathbb{C}$ is an eigenvalue of the matrix $B \in \mathbb{C}^{n \times n}$, we ask whether $B - zI$ is invertible. However, for matrices with an ill-conditioned spectrum, a small perturbation from B can change the answer from “yes” to “no”. Thus, asking whether $B - zI$ is invertible is not a robust question. Instead, we may ask if $\|(B - zI)^{-1}\|$ is large, where $\|\cdot\|$ denotes the matrix norm of interest [30].

This pattern of thinking leads to the analysis of the *pseudospectra* of B . For $\epsilon > 0$, the ϵ -*pseudospectrum* of B is the following set:

$$\text{Sp}_{\epsilon}(B) = \{z \in \mathbb{C} : \|(B - zI)^{-1}\|^{-1} \leq \epsilon\} = \text{Cl}(\{z \in \mathbb{C} : \|(B - zI)^{-1}\|^{-1} < \epsilon\}), \quad (2.8)$$

where $\text{Cl}(\cdot)$ denotes the closure of the inputted set [9]. Here, we adopt the convention that if $z \in \text{Sp}(B)$, then $\|(B - zI)^{-1}\|^{-1} = 0$, or $\|(B - zI)^{-1}\| = \infty$ [9, 30].

The ϵ -pseudospectrum of B can be equivalently defined as follows:

$$\text{Sp}_{\epsilon}(B) = \bigcup_{E \in \mathbb{C}^{n \times n}, \|E\| \leq \epsilon} \text{Sp}(B + E). \quad (2.9)$$

This second definition tells us that the ϵ -pseudospectrum of B is the region in the complex plane describing how far an ϵ -sized perturbation of B can move its eigenvalues [9]. In this sense, the pseudospectra of B describe the sensitivity of its eigenvalues to perturbations. The pseudospectra of B can therefore be used to identify the regions of the complex plane where eigenvalue computations are unreliable and where eigenvalues alone may provide an unreliable description of the underlying system.

Recall that B is a normal matrix when $B^*B = BB^*$, where B^* denotes the conjugate transpose of B . When B is a normal matrix, $\|(B - zI)^{-1}\|^{-1} = \text{dist}(z, \text{Sp}(B))$. In

this case, the ϵ -pseudospectrum of B consist of closed ϵ balls about each eigenvalue of B . Thus, an ϵ -sized perturbation of B cannot have eigenvalues that are further than an ϵ -distance from the spectrum of B . In this sense, the spectrum of B is well-conditioned.

In infinite dimensions, we consider a closed and densely defined linear operator $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$, on a Hilbert space $\mathcal{H}$. For each $\epsilon > 0$, the ϵ -pseudospectrum of A is the following set:²

$$\mathrm{Sp}_\epsilon(A) = \mathrm{Cl} \left(\{z \in \mathbb{C} : \|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} < \epsilon\} \right). \quad (2.10)$$

As before, we adopt the convention that if $z \in \mathrm{Sp}(A)$, then $\|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} = 0$, or $\|(A - zI)^{-1}\|_{\mathcal{H}} = \infty$ [9, 30]. Similarly to (2.9), the ϵ -pseudospectrum of A can be defined as follows [9, 30]:

$$\mathrm{Sp}_\epsilon(A) = \mathrm{Cl} \left(\bigcup_{E \in \mathcal{B}(\mathcal{H}), \|E\|_{\mathcal{H}} < \epsilon} \mathrm{Sp}(A + E) \right). \quad (2.11)$$

As is the case in finite dimensions, the pseudospectra of A can be used to explore the conditioning of the spectrum of A . When A is a normal operator, $\|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} = \mathrm{dist}(z, \mathrm{Sp}(A))$. In this case,

$$\mathrm{Sp}_\epsilon(A) = \mathrm{Sp}(A) + \mathfrak{B}_\epsilon(0), \quad (2.12)$$

where $\mathfrak{B}_\epsilon(0)$ denotes the closed ball of radius ϵ centered at zero [9]. Thus, a perturbation of A with a norm less than ϵ cannot have spectral values that are further than an ϵ -distance from the spectrum of A . In this sense, the spectrum of A is well-conditioned.

When A is not a normal operator, its ϵ -pseudospectrum may be much larger than $\mathrm{Sp}(A) + \mathfrak{B}_\epsilon(0)$. In this case, perturbations of A with norm less than ϵ may have spectral values that are much further than an ϵ -distance from the spectrum of A , indicating that the spectrum of A is ill-conditioned.

Before moving on, we note that pseudospectra have many applications beyond the analysis of the conditioning of spectra. For example, as we shall see, Colbrook's algorithm uses concepts related to pseudospectra to enable the reliable computation of spectra [9].

²Notice that in infinite dimensions, the equality in (2.8) need not hold. This is because in infinite dimensions, the resolvent norm of an unbounded operator on a Hilbert space can be constant on open sets [9].

2.2 Dangerous Pitfalls of the Discretize-Then-Solve Approach

The most common approach for computing the spectrum of an infinite-dimensional operator is the discretize-then-solve approach. This approach discretizes the infinite-dimensional operator with a finite-dimensional matrix and computes its eigenvalues. The success of this method relies on the hope that as the size of the discretization increases, the set of computed eigenvalues converges to the spectrum of the infinite-dimensional operator. While such an approach can be successful for many problems, it is known to fail catastrophically for others.

A common failure mode of the discretize-then-solve approach occurs when the computed spectrum contains values that do not belong to the spectrum of the infinite-dimensional operator. This phenomenon is referred to as spectral pollution, which is formally defined as follows:

Definition 2.2.1 (Spectral Pollution [9]). *Let $\Gamma_n(A) \subset \mathbb{C}$ denote an approximation of $\text{Sp}(A)$ for $n \in \mathbb{N}$. We say that the sequence of approximations $\{\Gamma_n(A)\}_{n \in \mathbb{N}}$ suffers from spectral pollution if there exists $\lambda \in \mathbb{C} \setminus \text{Sp}(A)$ such that*

$$\liminf_{n \rightarrow \infty} \text{dist}(\lambda, \Gamma_n(A)) = 0. \quad (2.13)$$

Spectral pollution is a more dangerous phenomenon than spurious eigenvalues of a fixed discretization. A spurious eigenvalue of a fixed discretization may move or disappear as the discretization is refined. However, spectral pollution can occur when there is a sequence of spurious eigenvalues that converge to a value not in the spectrum. This convergence is deceiving, as it makes the spurious limit seem like a genuine spectral value.

Spectral invisibility occurs when the discretization of the infinite-dimensional operator causes us to miss parts of its spectrum, even as the size of the discretization increases [9]. This failure mode is formally defined as follows:

Definition 2.2.2 (Spectral Invisibility [9]). *Let $\Gamma_n(A) \subset \mathbb{C}$ denote an approximation of $\text{Sp}(A)$ for $n \in \mathbb{N}$. We say that the sequence of approximations $\{\Gamma_n(A)\}_{n \in \mathbb{N}}$ suffers from spectral invisibility if there exists $\lambda \in \text{Sp}(A)$ such that*

$$\limsup_{n \rightarrow \infty} \text{dist}(\lambda, \Gamma_n(A)) > 0. \quad (2.14)$$

Even when the discretize-then-solve approach avoids both spectral pollution and spectral invisibility, there is a fundamental issue that arises when an infinite-dimensional

operator is discretized too quickly. For general operators, the *essential spectrum* refers to the $z \in \text{Sp}(A)$ for which $A - zI$ fails to be invertible in an ‘infinite-dimensional’ way³ [9]. The *discrete spectrum* refers to the $z \in \text{Sp}(A)$ for which $A - zI$ fails to be invertible in a ‘finite-dimensional’ way. Immediately discretizing A with a finite matrix results in a finite, discrete set of eigenvalues. As a result, the discretize-then-solve approach may struggle to capture the essential spectrum of an operator, and to distinguish it from the discrete spectrum.

2.2.1 The Maxwell Eigenproblem and Spectral Pollution

To illustrate the effect of these failure modes, we consider the Maxwell eigenproblem, a famous example for which naïve finite element discretizations exhibit spectral pollution [7].

The Maxwell eigenproblem describes electromagnetic oscillations in a resonator. The standard problem is posed on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^3$. The aim is to find the resonant frequencies $\omega \in \mathbb{R} \setminus \{0\}$ and the nonzero electromagnetic fields $(\mathbf{E}, \mathbf{H}) \in \mathbf{H}_0(\mathbf{curl}; \Omega) \times \mathbf{H}(\mathbf{curl}; \Omega)$ such that

$$\mathbf{curl} \mathbf{E} = i\omega \mathbf{H}, \quad \text{in } \Omega, \quad (2.15a)$$

$$\mathbf{curl} \mathbf{H} = -i\omega \mathbf{E}, \quad \text{in } \Omega, \quad (2.15b)$$

$$\mathbf{E} \times \mathbf{n} = 0, \quad \text{on } \partial\Omega. \quad (2.15c)$$

Here, $\mathbf{n}$ denotes the outward-facing normal, $\mathbf{H}(\mathbf{curl}; \Omega)$ contains the vectors in $L^2(\Omega)^3$ with $\mathbf{curl}$ in $L^2(\Omega)^3$, and $\mathbf{H}_0(\mathbf{curl}; \Omega)$ contains the elements of $\mathbf{H}(\mathbf{curl}; \Omega)$ that have zero tangential trace on $\partial\Omega$ [9].

The classical formulation of the eigenvalue problem is obtained by eliminating $\mathbf{H}$ and renaming $\mathbf{E}$ as the eigenfunction $\mathbf{u}$ in (2.15). The resulting eigenproblem consists of finding $\omega \in \mathbb{R} \setminus \{0\}$ and nonzero $\mathbf{u}$ such that

$$\mathbf{curl} \mathbf{curl} \mathbf{u} = \omega^2 \mathbf{u}, \quad \text{in } \Omega, \quad (2.16a)$$

$$\text{div} \mathbf{u} = 0, \quad \text{in } \Omega, \quad (2.16b)$$

$$\mathbf{u} \times \mathbf{n} = 0, \quad \text{on } \partial\Omega. \quad (2.16c)$$

The common variational formulation of (2.16) seeks $\omega \in \mathbb{R}$ and nonzero $\mathbf{u} \in \mathbf{H}_0(\mathbf{curl}; \Omega)$ such that

$$(\mathbf{curl} \mathbf{u}, \mathbf{curl} \mathbf{v})_{L^2(\Omega)} = \omega^2 (\mathbf{u}, \mathbf{v})_{L^2(\Omega)}, \quad \forall \mathbf{v} \in \mathbf{H}_0(\mathbf{curl}; \Omega), \quad (2.17)$$

³We omit a formal definition of the essential spectrum because there are at least five technical definitions for non-normal operators. The curious reader can consult [9, Ch. 8].

where $(\cdot, \cdot)_{L^2(\Omega)}$ denotes the $L^2(\Omega)$ inner product, i.e., $(\mathbf{u}, \mathbf{v})_{L^2(\Omega)} = \int_{\Omega} \mathbf{u} \cdot \bar{\mathbf{v}} d\mathbf{x}$. Notice that for nonzero values of ω , the eigenfunctions of (2.17) are also solutions of (2.16). However, at $\omega = 0$, (2.17) has an infinite-dimensional eigenspace containing $\mathbf{grad}(\mathbf{H}_0^1(\Omega))$ [7].

For simplicity, we consider the two-dimensional counterpart of (2.15), which shares the spectral pollution dangers of the three-dimensional problem. In particular, we set $\Omega = (0, \pi)^2$ and seek $\omega \in \mathbb{R}$ and nonzero $(\mathbf{E}, H) \in \mathbf{H}_0(\mathbf{rot}; \Omega) \times \mathbf{H}(\mathbf{rot}; \Omega)$ such that

$$\mathbf{rot} \mathbf{E} = i\omega H, \quad \text{in } \Omega, \quad (2.18a)$$

$$\mathbf{rot} H = -i\omega \mathbf{E}, \quad \text{in } \Omega, \quad (2.18b)$$

$$\mathbf{E} \cdot \mathbf{t} = 0, \quad \text{on } \partial\Omega. \quad (2.18c)$$

Here, $\mathbf{rot} \mathbf{E} = \partial_x E_2 - \partial_y E_1$, where $\mathbf{E} = (E_1, E_2)^T$, and $\mathbf{rot} H = (\partial_y H, -\partial_x H)^T$. Furthermore, we use $\mathbf{t}$ to denote the unit tangent to $\partial\Omega$, $\mathbf{H}(\mathbf{rot}; \Omega)$ to denote the space of functions in $L^2(\Omega)$ whose $\mathbf{rot}$ belongs in $L^2(\Omega)^2$, and $\mathbf{H}_0(\mathbf{rot}; \Omega)$ to denote the space of functions in $L^2(\Omega)^2$ whose $\mathbf{rot}$ belongs in $L^2(\Omega)$ [9].

It has been shown that the nonzero eigenvalues of (2.18) coincide with those of the Poisson eigenproblem with Neumann boundary conditions [7]. Consequently, the nonzero values of ω^2 are

$$\omega^2 = m^2 + n^2, \quad m, n \in \{0, 1, 2, \dots\}, \quad (m, n) \neq (0, 0). \quad (2.19)$$

To discretize (2.18) with the finite element method, we consider the two-dimensional counterpart of (2.17), that is, we seek $\omega \in \mathbb{R}$ and nonzero $\mathbf{u} \in \mathbf{H}_0(\mathbf{rot}; \Omega)$ such that

$$(\mathbf{rot} \mathbf{u}, \mathbf{rot} \mathbf{v})_{L^2(\Omega)} = \omega^2 (\mathbf{u}, \mathbf{v})_{L^2(\Omega)}, \quad \forall \mathbf{v} \in \mathbf{H}_0(\mathbf{rot}; \Omega), \quad (2.20)$$

where $\mathbf{H}_0(\mathbf{rot}; \Omega)$ denotes the elements of $\mathbf{H}(\mathbf{rot}; \Omega)$ that have zero tangential trace on $\partial\Omega$.

To transition from the infinite-dimensional setting to a finite-dimensional setting, we consider a sequence of subspaces $V_h \subset \mathbf{H}_0(\mathbf{rot}; \Omega)$. Then, we seek $\omega_h \in \mathbb{R}$ and nonzero $\mathbf{u}_h \in V_h$ such that

$$(\mathbf{rot} \mathbf{u}_h, \mathbf{rot} \mathbf{v}_h)_{L^2(\Omega)} = \omega_h^2 (\mathbf{u}_h, \mathbf{v}_h)_{L^2(\Omega)}, \quad \forall \mathbf{v}_h \in V_h. \quad (2.21)$$

Given a basis $V_h = \text{span}\{\phi_1, \dots, \phi_n\}$, (2.21) is equivalent to the following eigenproblem:

$$A_h \mathbf{x} = \omega_h^2 M_h \mathbf{x}, \quad (2.22)$$

where $\mathbf{x}$ is the vector of coefficients for $\mathbf{u}_h$, A_h is the stiffness matrix with entries $(A_h)_{ij} = (\text{rot } \phi_j, \text{rot } \phi_i)_{L^2(\Omega)}$, and M_h is the mass matrix with entries $(M_h)_{ij} = (\phi_j, \phi_i)_{L^2(\Omega)}$. Thus, from the perspective of linear algebra, the finite element discretization of an infinite-dimensional eigenvalue problem results in a generalized eigenproblem, where the mass matrix enforces the correct inner product [11]. In the current setting, M_h enforces the $L^2(\Omega)$ inner product, i.e., $(\mathbf{x}, \mathbf{y})_{L^2(\Omega)} = \mathbf{x}^* M_h \mathbf{y}$.

To compute solutions of eigenproblems such as (2.18), we employ Firedrake, a finite element software that enables users to specify variational problems directly, without requiring the user to construct the linear systems that result from discretization [17]. To use Firedrake to solve eigenproblems, we specify a variational formulation, such as (2.21), in Unified Form Language (UFL) [3]. Firedrake then assembles the resulting generalized eigenproblem and solves it using SLEPc [18].

To compute solutions of (2.18), we discretize $\Omega = (0, \pi)^2$ with the sequence of criss-cross meshes displayed in Figure 2.1. On each mesh, we take V_h to be the space of linear Lagrange elements satisfying the zero tangential trace boundary condition. With this choice, we write (2.21) in UFL and ask Firedrake to solve the resulting generalized eigenproblem using SLEPc's Krylov-Schur method with a shift-and-invert spectral transformation [18]. The results are displayed in Table 2.1.

Notice that Table 2.1 contains a sequence of eigenvalues converging to $\omega^2 = 6$, which is not an eigenvalue of the infinite-dimensional problem (2.18). Thus, this methodology suffers from spectral pollution. If we did not know the exact eigenvalues of (2.18), the apparently convergent sequence of eigenvalues could lead us to incorrectly conclude that $\omega^2 = 6$ is an eigenvalue of (2.18).

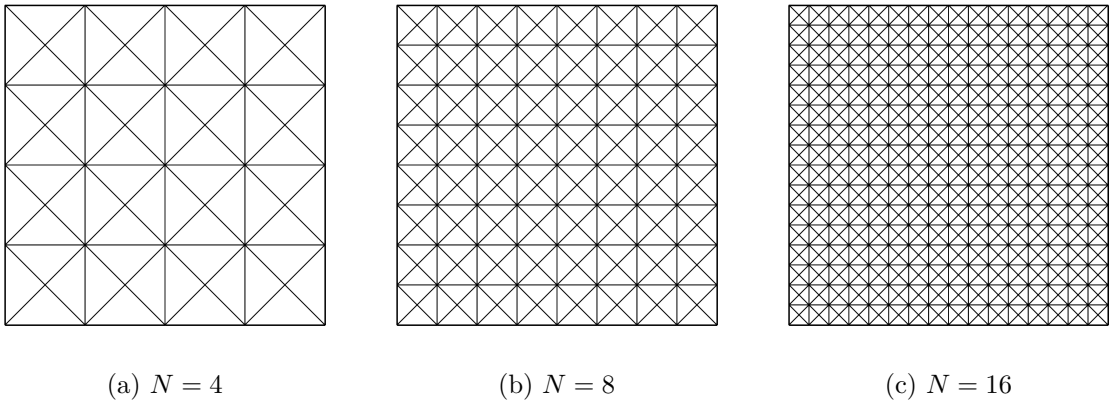


Figure 2.1: The first three meshes in the sequence of criss-cross meshes used to discretize (2.20) on $\Omega = (0, \pi)^2$. Here, N denotes the number of cells on each side of the mesh.

Exact ω^2	Computed ω_h^2				
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$
1	1.0170	1.0043	1.0011	1.0003	1.0001
1	1.0170	1.0043	1.0011	1.0003	1.0001
2	2.0678	2.0171	2.0043	2.0011	2.0003
4	4.2647	4.0680	4.0171	4.0043	4.0011
4	4.2647	4.0680	4.0171	4.0043	4.0011
5	5.3971	5.1063	5.0267	5.0067	5.0017
5	5.3971	5.1063	5.0267	5.0067	5.0017
6	5.6712	5.9229	5.9807	5.9952	5.9988
8	8.8141	8.2713	8.0685	8.0171	8.0043
9	10.2540	9.3408	9.0864	9.0217	9.0054
9	10.2540	9.3408	9.0864	9.0217	9.0054
DOF	62	254	1022	4094	16382

Table 2.1: Computed nonzero eigenvalues for the Maxwell eigenproblem on a sequence of criss-cross meshes. The red row indicates a sequence of spurious eigenvalues converging to $\omega^2 = 6$, which is not a valid eigenvalue of (2.18).

In general, for the Maxwell eigenproblem, whether spectral pollution occurs depends on the choice of mesh and finite elements [9]. For example, spectral pollution can be avoided if we employ edge, or Nédélec, elements as opposed to our current discretization. Edge elements get their name from the nature of their degrees of freedom, which, for the lowest-order approximation, are given by the moments along the edges of the triangulation of the mesh [7]. The success of edge elements for the Maxwell eigenproblem can be understood through finite element exterior calculus (FEEC) [9]. FEEC captures the essential geometric, topological, and algebraic structures of a problem at the discrete level. This linking of the continuous and discrete formulation of a problem enables the construction of stable and convergent finite element discretizations [4]. For the Maxwell eigenproblem, the success of edge elements can be explained by the fact that they fit within the framework of FEEC, while our current discretization does not [4, 7].

With this example in mind, it is clear that the pitfalls of the discretize-then-solve approach, such as spectral pollution, may render the approach unreliable for infinite-dimensional spectral computations. As a result, we turn to Matthew Colbrook’s alternative approach for computing the spectra of infinite-dimensional operators.

2.3 Colbrook’s Algorithm for Pseudospectra

The pitfalls of the discretize-then-solve approach motivate the development of alternative methods for computing the spectra of infinite-dimensional operators. In recent work, Matthew Colbrook has proposed an approach that avoids these pitfalls by working with the infinite-dimensional operator as far as possible, thereby delaying discretization [9]. In this way, Colbrook shows that when and how discretization is introduced is crucial to the reliable computation of spectra.

In this section, we introduce Colbrook’s algorithm for computing pseudospectra. We focus on his algorithm for pseudospectral computations rather than his algorithm for spectral computations for three main reasons. First, it applies to a much broader class of operators than his algorithm for spectra. Second, it forms the main computational component of his algorithm for computing spectra. Finally, as discussed in Section 2.1.2, the pseudospectra of an operator are often more informative than the spectrum alone.

In the first subsection, we introduce the fundamental tool for computing pseudospectra, namely, the injection modulus of an operator. In the second subsection, we use this tool to construct Colbrook’s algorithm for pseudospectra. Then, in the third subsection, we specialize the algorithm to the finite element setting. Finally, in the final subsection, we demonstrate the success of Colbrook’s algorithm by applying it to the Maxwell eigenproblem.

2.3.1 Characterizing the Spectrum with Injection Moduli

Let $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$ be a closed and densely defined linear operator on the Hilbert space $\mathcal{H}$. The first ingredient for Colbrook’s algorithm is the *injection modulus* of A , which is defined as

$$\sigma_{\inf}(A) = \inf\{\|Ax\|_{\mathcal{H}} : x \in D(A), \|x\|_{\mathcal{H}} = 1\}. \quad (2.23)$$

The injection modulus is a quantitative measure of ‘how injective’ an operator is [9]. In this way, it can be viewed as a generalization of the smallest singular value of a finite-dimensional matrix. However, we must be careful not to confuse the injection modulus with the smallest singular value, since infinite-dimensional operators can exhibit behaviours that are quite different from those of finite-dimensional operators [9]. For example, an infinite-dimensional operator A can be injective but have $\sigma_{\inf}(A) = 0$. This may happen when there is a sequence of unit vectors, $\{x_n\}_{n \in \mathbb{N}}$, such that $\|Ax_n\|_{\mathcal{H}} \rightarrow 0$ but there is no nonzero $x \in D(A)$ that satisfies $\|Ax\|_{\mathcal{H}} = 0$.

One of the most fundamental tools in Colbrook's algorithm is the following function of $z \in \mathbb{C}$,

$$\gamma(z, A) = \min \{ \sigma_{\inf}(A - zI), \sigma_{\inf}(A^* - \bar{z}I) \}. \quad (2.24)$$

With this, Colbrook characterizes the resolvent norm $\|(A - zI)^{-1}\|_{\mathcal{H}}$ in the following way:

Lemma 2.3.1 (Injection Moduli and Resolvent Norms [9]). *Let A be a closed and densely defined operator on a Hilbert space $\mathcal{H}$. We have that*

$$\gamma(z, A) = \|(A - zI)^{-1}\|_{\mathcal{H}}^{-1}, \quad (2.25)$$

where we adopt the convention that $\|(A - zI)^{-1}\|_{\mathcal{H}}^{-1} = 0$ if $z \in \text{Sp}(A)$. Furthermore, if $z \notin \text{Sp}(A)$, then $\sigma_{\inf}(A - zI) = \sigma_{\inf}(A^* - \bar{z}I)$.

In finite dimensions, we have that $\sigma_{\inf}(A - zI) = \sigma_{\inf}(A^* - \bar{z}I)$ for every $z \in \mathbb{C}$. However, in infinite dimensions, it is possible to have $z \in \text{Sp}(A)$ and $\sigma_{\inf}(A - zI) \neq \sigma_{\inf}(A^* - \bar{z}I)$. To see this, consider the right-shift operator as defined in (1.3), i.e., the operator $A : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ defined by $A(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$. This operator satisfies $\|Ax\|_{\ell^2(\mathbb{N})} = \|x\|_{\ell^2(\mathbb{N})}$. Thus, $\sigma_{\inf}(A) = 1$. Now consider its Hilbert adjoint A^* , which is the left-shift operator $A^*(x_1, x_2, \dots) = (x_2, x_3, \dots)$. Notice that $A^*(1, 0, \dots) = (0, 0, \dots)$. Hence, $\sigma_{\inf}(A) \neq \sigma_{\inf}(A^*) = 0$.

With Lemma 2.3.1 in mind, we can define the spectrum of A as follows:

$$\text{Sp}(A) = \{z \in \mathbb{C} : \gamma(z, A) = 0\}. \quad (2.26)$$

Similarly, we can define the ϵ -pseudospectrum as the following set:

$$\text{Sp}_{\epsilon}(A) = \text{Cl}(\{z \in \mathbb{C} : \gamma(z, A) < \epsilon\}). \quad (2.27)$$

Thus, to approximate $\text{Sp}(A)$ and $\text{Sp}_{\epsilon}(A)$, we can approximate $\gamma(z, A)$. This idea underpins Colbrook's algorithms for computing the spectra and pseudospectra of infinite-dimensional operators. As discussed, we focus on Colbrook's algorithm for computing pseudospectra because it applies to a broad class of operators, because it forms the main computational component of his algorithm for computing spectra, and because pseudospectral computations tend to be more interesting than spectral computations.

2.3.2 Computing Pseudospectra with Operator Folding

Colbrook's algorithm for computing the pseudospectra of an operator A relies on the characterization of $\text{Sp}_{\epsilon}(A)$ using $\gamma(z, A)$ in (2.27). The key idea is to approximate

$\gamma(z, A)$ at chosen test points $z \in \mathbb{C}$ and retain all points for which $\gamma(z, A) < \epsilon$. To construct this methodology, Colbrook makes the following assumption:

Assumption 2.3.2 (Required Convergence of $\Phi_n(z, A)$ [9]). *To compute $\text{Sp}_\epsilon(A)$, assume that we can compute functions $\Phi_n(z, A)$ such that*

$$\lim_{n \rightarrow \infty} \sup_{z \in \mathcal{C}} |\Phi_n(z, A) - \gamma(z, A)| = 0 \quad \forall \text{ compact } \mathcal{C} \subset \mathbb{C}, \quad (2.28)$$

where the speed of convergence is allowed to depend on A and $\mathcal{C}$. Furthermore, we assume that $\gamma(z, A) \leq \Phi_n(z, A)$ for sufficiently large n , depending on A but not z .

To satisfy this assumption, we introduce *operator folding*. Suppose that A is a closed and densely defined operator on a separable Hilbert space $\mathcal{H}$, i.e., a Hilbert space that contains a countable orthonormal basis [9]. Let $\{\mathcal{P}_n\}$ be orthogonal projections such that $\mathcal{P}_n^* \mathcal{P}_n$ strongly converges to the identity as $n \rightarrow \infty$ [9]. The key idea behind operator folding is to approximate $\gamma(z, A)$ using the following relation:

$$\sigma_{\inf}((A - zI)\mathcal{P}_n^*) = \sqrt{\sigma_{\inf}(\mathcal{P}_n(A - zI)^*(A - zI)\mathcal{P}_n^*)}. \quad (2.29)$$

That is, we approximate $\gamma(z, A)$ with

$$\Phi_n(z, A) = \min \{ \sigma_{\inf}((A - zI)\mathcal{P}_n^*), \sigma_{\inf}((A - zI)^*\mathcal{P}_n^*) \}, \quad (2.30)$$

which satisfies Assumption 2.3.2 under suitable conditions on $D(A)$ and $D(A^*)$ [9, 21].

To compute $\Phi_n(z, A)$ in (2.30), we discretize variational problems for $\gamma(z, A)$. In particular, we consider the following quadratic forms:

$$Q_{z,A}^{(1)}(u, v) = ((A - zI)u, (A - zI)v)_{\mathcal{H}}, \quad u, v \in D(A), \quad (2.31)$$

$$Q_{z,A}^{(2)}(u, v) = ((A - zI)^*u, (A - zI)^*v)_{\mathcal{H}}, \quad u, v \in D(A^*), \quad (2.32)$$

which induce the Rayleigh quotients

$$\mathcal{R}_{z,A}^{(1)}(u) = \frac{Q_{z,A}^{(1)}(u, u)}{(u, u)_{\mathcal{H}}} = \frac{\|(A - zI)u\|_{\mathcal{H}}^2}{\|u\|_{\mathcal{H}}^2}, \quad (2.33)$$

$$\mathcal{R}_{z,A}^{(2)}(u) = \frac{Q_{z,A}^{(2)}(u, u)}{(u, u)_{\mathcal{H}}} = \frac{\|(A - zI)^*u\|_{\mathcal{H}}^2}{\|u\|_{\mathcal{H}}^2}. \quad (2.34)$$

These Rayleigh quotients can be used to characterize the injection modulus of $A - zI$ and $(A - zI)^*$ in the following way:

$$\sigma_{\inf}(A - zI) = \sqrt{\mu^{(1)}(z)}, \quad \mu^{(1)}(z) = \inf_{u \in D(A) \setminus \{0\}} \mathcal{R}_{z,A}^{(1)}(u), \quad (2.35)$$

$$\sigma_{\inf}(A^* - \bar{z}I) = \sqrt{\mu^{(2)}(z)}, \quad \mu^{(2)}(z) = \inf_{u \in D(A^*) \setminus \{0\}} \mathcal{R}_{z,A}^{(2)}(u). \quad (2.36)$$

With this, $\gamma(z, A)$ can be written as

$$\gamma(z, A) = \sqrt{\min \{\mu^{(1)}(z), \mu^{(2)}(z)\}}. \quad (2.37)$$

Consequently, computing $\gamma(z, A)$ is equivalent to computing the bottom of the spectrum of the two folded operators $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$.

It is at this stage that discretization enters Colbrook's methodology. To compute $\Phi_n(z, A)$, we discretize the minimization problems in (2.37) and approximate the bottom of the spectrum of $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$. Letting $\mu_n^{(1)}(z)$ and $\mu_n^{(2)}(z)$ denote the computed approximations of $\mu^{(1)}(z)$ and $\mu^{(2)}(z)$, respectively, we set

$$\Phi_n(z, A) = \sqrt{\min \{\mu_n^{(1)}(z), \mu_n^{(2)}(z)\}}. \quad (2.38)$$

With approximation (2.38) at hand, we can formulate Colbrook's algorithm for computing the ϵ -pseudospectrum of A . First, we select a collection of test points in the complex plane, denoted $\text{Grid} \subset \mathbb{C}$. Then, we iterate over each $z \in \text{Grid}$, compute $\Phi_n(z, A)$, and approximate $\text{Sp}_\epsilon(A)$ by the following set of points:

$$\Gamma_n^\epsilon(A) = \{z \in \text{Grid} : \Phi_n(z, A) < \epsilon\}. \quad (2.39)$$

This method is outlined in Algorithm 2. Under Assumption 2.3.2, we have that

$$\Gamma_n^\epsilon(A) \subset \{z \in \mathbb{C} : \gamma(z, A) < \epsilon\} \subset \text{Sp}_\epsilon(A). \quad (2.40)$$

Furthermore, Colbrook shows that the output of Algorithm 1, i.e., $\Gamma_n^\epsilon(A)$, converges to $\text{Sp}_\epsilon(A)$ as the chosen Grid is extended to approximate all of $\mathbb{C}$ and as $n \rightarrow \infty$ [9].

Algorithm 1 Colbrook's Algorithm for Computing $\text{Sp}_\epsilon(A)$.

Input: $\text{Grid} \subset \mathbb{C}$, $\epsilon > 0$.

- 1: For each $z \in \text{Grid}$, compute $\mu_n^{(1)}(z)$ and $\mu_n^{(2)}(z)$.
- 2: Set $\Phi_n(z, A) = \sqrt{\min \{\mu_n^{(1)}(z), \mu_n^{(2)}(z)\}}$.
- 3: Set $\Gamma_n^\epsilon(A) = \{z \in \text{Grid} : \Phi_n(z, A) < \epsilon\}$.

Output: $\Gamma_n^\epsilon(A)$, an approximation of $\text{Sp}_\epsilon(A)$.

The computation of $\mu_n^{(1)}(z)$ and $\mu_n^{(2)}(z)$ needed for Algorithm 1 depends on the chosen discretization. In the next subsection, we employ the finite element method to compute $\mu_n^{(1)}(z)$ and $\mu_n^{(2)}(z)$ as the smallest eigenvalues of two auxiliary eigenproblems.

2.3.3 Operator Folding with the Finite Element Method

Colbrook's formulation of $\gamma(z, A)$ in (2.37) considers a closed and densely defined operator $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$. To approximate $\gamma(z, A)$ by $\Phi_n(z, A)$ in (2.38), we discretize the minimization problems in (2.37). To do so, we employ the finite element method.

The finite element method is often formulated for operators of the form $L : V \rightarrow V^*$, where V^* is the dual, or anti-dual, of V . This differs from the setting of Colbrook's algorithm, which is formulated for densely defined operators. To see this discrepancy, consider the Dirichlet Laplacian on $\mathcal{H} = L^2(\Omega)$, defined as

$$A := -\Delta, \quad D(A) = \{u \in H_0^1(\Omega) : -\Delta u \in L^2(\Omega)\}. \quad (2.41)$$

This operator fits into the setting of Colbrook's algorithm because it maps its domain, which is dense in the Hilbert space $\mathcal{H}$, into $\mathcal{H}$. When employing the finite element method, we consider the 'weak' Laplacian $L : V \rightarrow V^*$, where $V := H_0^1(\Omega)$. This operator is defined as follows:

$$\langle Lu, v \rangle_{V^*, V} := (\nabla u, \nabla v)_{L^2(\Omega)}, \quad \forall u, v \in V, \quad (2.42)$$

where $\langle \cdot, \cdot \rangle_{V^*, V}$ denotes the action of an element of V^* on an element of V . Unlike the operator A , the operator L is not a densely defined operator on $\mathcal{H}$. Thus, L does not directly fit the setting of Colbrook's algorithm.

Motivated by this discrepancy, we develop a finite element formulation of Colbrook's algorithm. We begin with Colbrook's densely defined operator A and its Hilbert adjoint A^* . Furthermore, we define $V := D(A)$ and $W := D(A^*)$. To retain the metric that defines $\gamma(z, A)$, we introduce the *inverse Riesz map* $M : \mathcal{H} \rightarrow \mathcal{H}^*$, defined by

$$\langle Mu, v \rangle_{\mathcal{H}^*, \mathcal{H}} = (u, v)_{\mathcal{H}} \quad \forall u, v \in \mathcal{H}. \quad (2.43)$$

For ease of notation, we use $m(u, v)$ to denote the sesquilinear form in (2.43).

Using M , we introduce the operator $L := MA : V \rightarrow \mathcal{H}^*$, defined by

$$\langle Lu, v \rangle_{\mathcal{H}^*, \mathcal{H}} = (Au, v)_{\mathcal{H}}, \quad \forall u \in V, \forall v \in \mathcal{H}. \quad (2.44)$$

Furthermore, using M^{-1} , which is the Riesz map from $\mathcal{H}^*$ to $\mathcal{H}$, we write $A = M^{-1}L$. In doing so, we have that

$$M^{-1}(L - zM) = A - zI. \quad (2.45)$$

To represent Colbrook's $\gamma(z, A)$ in our desired setting, we introduce the *Banach adjoint*. Consider a bounded linear operator $T : X \rightarrow Y$, where X and Y are two

Banach spaces. The Banach adjoint of T is the bounded operator $T^\dagger : Y^* \rightarrow X^*$ that satisfies

$$\langle T^\dagger f, u \rangle_{X^*, X} = \langle f, Tu \rangle_{Y^*, Y}, \quad \forall f \in Y^*, \forall u \in X. \quad (2.46)$$

We also introduce the *canonical injection* $J_{\mathcal{H}} : \mathcal{H} \rightarrow \mathcal{H}^{**}$. In the real case, $J_{\mathcal{H}}$ satisfies

$$\langle J_{\mathcal{H}} u, f \rangle_{\mathcal{H}^{**}, \mathcal{H}^*} = \langle f, u \rangle_{\mathcal{H}^*, \mathcal{H}} \quad \forall f \in \mathcal{H}^*, \forall u \in \mathcal{H}. \quad (2.47)$$

In the complex case, $J_{\mathcal{H}}$ satisfies

$$\langle J_{\mathcal{H}} u, f \rangle_{\mathcal{H}^{**}, \mathcal{H}^*} = \overline{\langle f, u \rangle_{\mathcal{H}^*, \mathcal{H}}} \quad \forall f \in \mathcal{H}^*, \forall u \in \mathcal{H}. \quad (2.48)$$

With these two new operators, we use (2.45) to write the quadratic form in (2.31) as the following sesquilinear form:

$$Q_{z,A}^{(1)}(u, v) = ((A - zI)u, (A - zI)v)_{\mathcal{H}} \quad (2.49)$$

$$= (M^{-1}(L - zM)u, M^{-1}(L - zM)v)_{\mathcal{H}} \quad (2.50)$$

$$= \langle (L - zM)u, M^{-1}(L - zM)v \rangle_{\mathcal{H}^*, \mathcal{H}} \quad (2.51)$$

$$= \langle (M^{-1}(L - zM))^\dagger (L - zM)u, v \rangle_{V^*, V} \quad (2.52)$$

$$= \langle (L - zM)^\dagger (M^{-1})^\dagger (L - zM)u, v \rangle_{V^*, V} \quad (2.53)$$

$$= \langle (L - zM)^\dagger J_{\mathcal{H}} M^{-1}(L - zM)u, v \rangle_{V^*, V}, \quad (2.54)$$

where (2.53) follows from the fact that $(M^{-1}(L - zM))^\dagger = (L - zM)^\dagger (M^{-1})^\dagger$, and (2.54) follows from the definition of $J_{\mathcal{H}}$ and $(M^{-1})^\dagger$.

With this result, for any $u \in V \setminus \{0\}$, the Rayleigh quotient in (2.33) can be written as

$$\mathcal{R}_{z,A}^{(1)}(u) = \frac{Q_{z,A}^{(1)}(u, u)}{m(u, u)} = \frac{\langle (L - zM)^\dagger J_{\mathcal{H}} M^{-1}(L - zM)u, u \rangle_{V^*, V}}{m(u, u)}. \quad (2.55)$$

To obtain an analogous formulation of the Rayleigh quotient in (2.34), we consider the operator $S : W \rightarrow \mathcal{H}^*$ that satisfies $S = MA^*$. Since

$$M^{-1}(S - \bar{z}M) = (A - zI)^*, \quad (2.56)$$

we can follow the same procedure used to obtain (2.55). In doing so, we have that for any $u \in W \setminus \{0\}$,

$$\mathcal{R}_{z,A}^{(2)}(u) = \frac{Q_{z,A}^{(2)}(u, u)}{m(u, u)} = \frac{\langle (S - \bar{z}M)^\dagger J_{\mathcal{H}} M^{-1}(S - \bar{z}M)u, u \rangle_{W^*, W}}{m(u, u)}. \quad (2.57)$$

With these new formulations of $\mathcal{R}_{z,A}^{(1)}(u)$ and $\mathcal{R}_{z,A}^{(2)}(u)$, we can write $\gamma(z, A)$ as

$$\gamma(z, A) = \sqrt{\min \{\mu^{(1)}(z), \mu^{(2)}(z)\}}, \quad (2.58)$$

where

$$\mu^{(1)}(z) = \inf_{u \in V \setminus \{0\}} \mathcal{R}_{z,A}^{(1)}(u) = \inf_{u \in V \setminus \{0\}} \frac{\langle (L - zM)^\dagger J_{\mathcal{H}} M^{-1} (L - zM) u, u \rangle_{V^*, V}}{m(u, u)}, \quad (2.59)$$

$$\mu^{(2)}(z) = \inf_{u \in W \setminus \{0\}} \mathcal{R}_{z,A}^{(2)}(u) = \inf_{u \in W \setminus \{0\}} \frac{\langle (S - \bar{z}M)^\dagger J_{\mathcal{H}} M^{-1} (S - \bar{z}M) u, u \rangle_{W^*, W}}{m(u, u)}. \quad (2.60)$$

To approximate $\mu^{(1)}(z)$, we consider a finite element space $V_h \subset V$ and write

$$\mu_h^{(1)}(z) := \min_{u_h \in V_h \setminus \{0\}} \frac{Q_{z,A}^{(1)}(u_h, u_h)}{m(u_h, u_h)}. \quad (2.61)$$

Thus, $\mu_h^{(1)}(z)$ can be obtained from the smallest eigenvalue of the following eigenproblem: find $\lambda_h \in \mathbb{R}$ and nonzero $u_h \in V_h$ such that

$$Q_{z,A}^{(1)}(u_h, v_h) = \lambda_h m(u_h, v_h) \quad \forall v_h \in V_h. \quad (2.62)$$

Similarly, to approximate $\mu^{(2)}(z)$, we introduce the finite element space $W_h \subset W$ and write

$$\mu_h^{(2)}(z) := \min_{u_h \in W_h \setminus \{0\}} \frac{Q_{z,A}^{(2)}(u_h, u_h)}{m(u_h, u_h)}. \quad (2.63)$$

Thus, $\mu_h^{(2)}(z)$ can be obtained from the smallest eigenvalue of the following eigenproblem: find $\lambda_h \in \mathbb{R}$ and nonzero $u_h \in W_h$ such that

$$Q_{z,A}^{(2)}(u_h, w_h) = \lambda_h m(u_h, w_h) \quad \forall w_h \in W_h. \quad (2.64)$$

The advantage of formulating our approximations in this way is that the sesquilinear representations of $Q_{z,A}^{(1)}(u_h, v_h)$, $Q_{z,A}^{(2)}(u_h, v_h)$, and $m(u_h, v_h)$ can be assembled directly in Firedrake. As a result, we can employ Firedrake's solvers to compute the smallest eigenvalues, $\mu_h^{(1)}(z)$ and $\mu_h^{(2)}(z)$, of the auxiliary eigenproblems, (2.62) and (2.64), respectively. In doing so, our approximation of $\gamma(z, A)$ is

$$\Phi_h(z, A) = \sqrt{\min \{\mu_h^{(1)}(z), \mu_h^{(2)}(z)\}}. \quad (2.65)$$

Here, we use the subscript h , rather than Colbrook's index n in Algorithm 1, to emphasize that the approximation of $\gamma(z, A)$ is obtained using the finite element method. Thus, $\Phi_h(z, A)$ is to be viewed as the finite element realization of Colbrook's $\Phi_n(z, A)$.

Algorithm 2 Finite Element Formulation of Algorithm 1.

Input: $\text{Grid} \subset \mathbb{C}$, $\epsilon > 0$, $V_h \subset D(A)$ and $W_h \subset D(A^*)$.

- 1: For each $z \in \text{Grid}$, compute $\mu_h^{(1)}(z)$ as the smallest eigenvalue of (2.62).
- 2: For each $z \in \text{Grid}$, compute $\mu_h^{(2)}(z)$ as the smallest eigenvalue of (2.64).
- 3: Set $\Phi_h(z, A) = \sqrt{\min \left\{ \mu_h^{(1)}(z), \mu_h^{(2)}(z) \right\}}$.
- 4: Set $\Gamma_h^\epsilon(A) = \{z \in \text{Grid} : \Phi_h(z, A) < \epsilon\}$.

Output: $\Gamma_h^\epsilon(A)$, a finite element approximation of $\text{Sp}_\epsilon(A)$.

In this new setting, the limit $h \rightarrow 0$ plays the role of $n \rightarrow \infty$ in Colbrook's abstract formulation. Following this line of thinking, Algorithm 2 reformulates Colbrook's algorithm, Algorithm 1, in our finite element setting.

Before moving on, we emphasize that our finite element formulation of Colbrook's algorithm applies to operators, $L : V \rightarrow \mathcal{H}^*$ and $S : W \rightarrow \mathcal{H}^*$, rather than to the more general operators, $L : V \rightarrow V^*$ and $S : W \rightarrow W^*$. This restriction allows us to work with the Riesz map $M : \mathcal{H} \rightarrow \mathcal{H}^*$ and its inverse $M^{-1} : \mathcal{H}^* \rightarrow \mathcal{H}$. As a result, our formulation of Colbrook's algorithm is limited to densely defined operators on $\mathcal{H}$ whose variational realizations map into $\mathcal{H}^*$. All of the operators considered in this work satisfy this requirement. However, many standard operators do not. For example, the 'weak' Laplacian in (2.41) does not satisfy this requirement because it maps into $(H_0^1(\Omega))^*$, whose elements need not belong to $(L^2(\Omega))^*$. Extending our current formulation of Colbrook's algorithm so that it can be applied to a broader class of operators is a topic of future work.

2.3.4 Application to the Maxwell Eigenproblem

To demonstrate the success of Colbrook's algorithm, we apply it to the Maxwell eigenproblem (2.18) of Section 2.2.1. To do so, we consider the closed and densely defined operator $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$, where

$$A(\mathbf{E}, H) = (i\text{rot}H, -i\text{rot}\mathbf{E}). \quad (2.66)$$

Here, $\Omega = (0, \pi)^2$, $\mathcal{H} = L^2(\Omega)^2 \times L^2(\Omega)$, and $D(A) = \mathbf{H}_0(\text{rot}; \Omega) \times \mathbf{H}(\text{rot}; \Omega)$.

To apply our formulation of Colbrook's algorithm, i.e., Algorithm 2, we first recognize that A is self-adjoint. As a result, $\text{Sp}(A) \subset \mathbb{R}$ and $\gamma(z, A) = \text{dist}(z, \text{Sp}(A))$. Furthermore, since A is self-adjoint, we only consider a single folded operator, $(A - zI)^*(A - zI)$, and its corresponding auxiliary eigenproblem.

To formulate the auxiliary eigenproblem used to compute $\Phi_h(z, A)$ at each $z \in$

Grid $\subset \mathbb{C}$, we follow the discussion of Section 2.3.3 and define

$$Q_{z,A}(u, v) := ((A - zI)u, (A - zI)v)_{\mathcal{H}}. \quad (2.67)$$

With this,

$$\gamma^2(z, A) = \text{dist}^2(z, \text{Sp}(A)) = \inf_{u \in D(A) \setminus \{0\}} \frac{Q_{z,A}(u, u)}{m(u, u)}. \quad (2.68)$$

To discretize, we restrict ourselves to the finite-element space $V_h \subset D(A)$. We then solve the auxiliary eigenproblem

$$Q_{z,A}(u_h, v_h) = \lambda_h m(u_h, v_h) \quad \forall v_h \in V_h, \quad (2.69)$$

and denote its smallest eigenvalue by $\mu_h(z)$. With this, our approximate bound on the distance to the spectrum is

$$\Phi_h(z, A) = \sqrt{\mu_h(z)}. \quad (2.70)$$

Thus, to implement Algorithm 2, we iterate over $z \in \text{Grid} \subset \mathbb{C}$, compute the smallest eigenvalue of (2.69), and approximate the bound on the distance to the spectrum using (2.70).

To demonstrate the success of this approach, we consider structured triangular meshes with $N = 32$ and $N = 128$ cells along each side of the domain. As in Section 2.2.1, we take V_h to be the space of linear Lagrange elements that satisfy the required zero-tangential trace boundary condition. Since $\text{Sp}(A) \subset \mathbb{R}$, we take $\text{Grid} \subset \mathbb{R}$. In particular, we set

$$\text{Grid} = \{0.01j : j = 1, 2, \dots, 400\}. \quad (2.71)$$

For each $z \in \text{Grid}$, we assemble (2.69) in Firedrake and compute the smallest eigenvalue using SLEPc's Krylov-Schur method with a shift-and-invert spectral transformation [17, 18]. In doing so, we compute $\Phi_h(z, A)$ as in (2.70). The results are displayed in Figure 2.2.

Figure 2.2 shows that Algorithm 2 does not approximate the spurious spectral value $\sqrt{6} \approx 2.45$. Thus, despite our choice of linear Lagrange elements, Algorithm 2 avoids the spectral pollution exhibited by the discretize-then-solve approach in Section 2.2.1. However, notice that for coarse discretizations, the computed bound on the distance to the spectrum is poor. As a result, in Figure 2.2, we employ uniform mesh refinement to increase the accuracy of the bound.

However, extensive uniform refinement at each $z \in \text{Grid}$ can make Algorithm 2 computationally expensive. In an effort to efficiently compute accurate bounds on

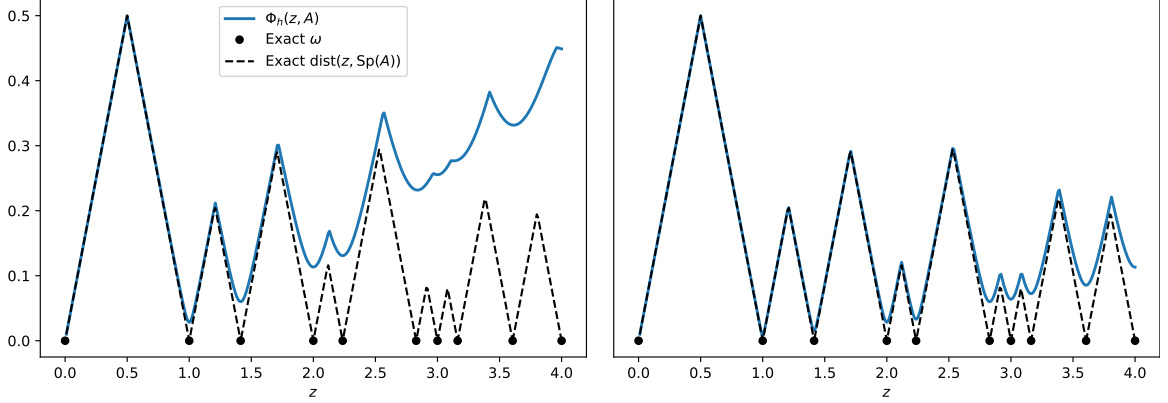


Figure 2.2: The results from applying Algorithm 2 to the two-dimensional Maxwell operator (2.66). The blue curve denotes $\Phi_h(z, A)$, the black dots denote the exact values of ω , and the black dashed curve denotes $\text{dist}(z, \text{Sp}(A))$. On the left, $N = 32$, on the right, $N = 128$.

the distance to the spectrum, we propose coupling Algorithm 2 with an adaptive mesh refinement technique driven by the Dual-Weighted Residual (DWR) method for symmetric eigenproblems. We discuss this methodology next.

2.4 The Dual-Weighted Residual (DWR) Method for Symmetric Eigenproblems

In this section, we provide an overview of the Dual-Weighted Residual (DWR) method that we apply to the auxiliary eigenproblems in Algorithm 2. In the first subsection, we discuss the role of *a posteriori* error estimates in constructing adaptive finite element methods. In the subsections that follow, we introduce the DWR error estimate for symmetric eigenproblems and discuss its localization for adaptive mesh refinement.

2.4.1 Adaptive Finite Element Methods

Since the finite element methodology provides an approximation to the solution of an underlying infinite-dimensional variational problem, we may want to measure, control, and effectively minimize the approximation error [2]. To do so, we turn to *a posteriori* error estimates. Given a computed solution, *a posteriori* error estimates provide an estimate of the error in that particular approximation [1]. As a result, *a posteriori* error estimates are used to determine which parts of the current discretization should be modified to reduce the error in the current approximation [1].

This idea gives rise to adaptive finite-element methods (AFEMs), which are techniques for choosing a finite-dimensional space, V_h , that is optimal with respect to computational complexity and accuracy [1]. This is typically done by adaptively re-

fining the mesh used to construct V_h in regions where the error estimate is large, giving rise to adaptive mesh refinement techniques (AMRs) [1]. These adaptive mesh refinement techniques are particularly useful when the error estimate is large only in certain regions of the domain. In this case, choosing a fine uniform mesh may be a waste of computational effort. In contrast, adaptive mesh techniques refine the mesh only in regions where the estimated error is large. For this reason, AFEMs, that take the form of AMRs, are useful tools for improving the efficiency and accuracy of finite element approximations.

When applying the finite element methodology to solve a problem arising in application, scientists and engineers are often interested in a derived quantity of interest, rather than in the solution field over the whole domain [14]. For example, when solving a problem in fluid dynamics, we may be interested in the vorticity of an eddy, rather than the velocity field [1]. In the case of Algorithm 2, our quantity of interest is the smallest eigenvalue of each auxiliary eigenproblem rather than its eigenfunction.

Following this line of thinking, it may be more beneficial to compute and control error estimates in a prescribed quantity of interest rather than in the solution field itself. This idea gives rise to the Dual-Weighted Residual (DWR) method, which provides *a posteriori* error estimates for prescribed quantities of interest.

2.4.2 The DWR Method for the Eigenvalue Error

To illustrate the ideas underlying the DWR method, we begin in an abstract setting. Let u denote the solution of the underlying infinite-dimensional problem, often referred to as the *primal problem*. Let $J(\cdot)$ be a scalar function used to compute the quantity of interest $J(u)$. When we discretize the primal problem with the finite element method, we obtain a solution approximation, u_h , and the corresponding approximation of the quantity of interest, $J(u_h)$. The DWR method aims to both estimate and control the error in the computed quantity of interest,

$$|J(u) - J(u_h)|. \quad (2.72)$$

To estimate this error, the DWR method embeds the given problem into the framework of optimal control [6]. To do so, the method introduces an associated *dual problem* that depends on the chosen quantity of interest. Using the solution of the dual problem, the method provides an error estimate in the form of a primal residual weighted by a ‘sensitivity parameter’ obtained from the dual solution. In this way, the dual solution is used to describe sensitivity of the quantity of interest to errors in the computed primal solution. This allows for mesh adaptation driven by the error

in the computed quantity of interest rather than by the error in the computed primal solution [6].

We now specialize the DWR framework to eigenproblems, taking the eigenvalue itself as the chosen quantity of interest. To do so, we consider a subspace V of a Hilbert space $\mathcal{H}$, where the embedding $V \hookrightarrow \mathcal{H}$ is compact⁴. Let $\mathcal{A}$ and $\mathcal{M}$ be linear operators on V . We seek a $\lambda \in \mathbb{C}$ and nonzero $u \in V$ such that

$$\mathcal{A}u = \lambda \mathcal{M}u, \quad (2.73)$$

where $\mathcal{M}$ is assumed to be self-adjoint and positive semidefinite [5].

As per usual, we prefer the variational formulation

$$a(u, v) = \lambda m(u, v) \quad \forall v \in V, \quad (2.74)$$

where $a(\cdot, \cdot)$ and $m(\cdot, \cdot)$ are the sesquilinear forms induced by $\mathcal{A}$ and $\mathcal{M}$, respectively. Furthermore, all eigenfunctions that satisfy (2.74) are assumed to be normalized by $m(u, u) = 1$.

To discretize this eigenproblem, we introduce the finite-dimensional subspace $V_h \subset V$ and seek $\lambda_h \in \mathbb{C}$ and nonzero $u_h \in V_h$ that satisfy

$$a(u_h, v_h) = \lambda_h m(u_h, v_h) \quad \forall v_h \in V_h, \quad (2.75)$$

where it is assumed that $m(u_h, u_h) = 1$. With this, our goal is to control the error in λ_h . In other words, the quantity of interest is $J(u_h) = \lambda_h$.

As previously discussed, the DWR method introduces a dual problem whose solution is used to describe the sensitivity of the quantity of interest to errors in the primal solution. For a general eigenvalue problem, the dual problem takes the form of the adjoint eigenproblem:

$$\mathcal{A}^*u^* = \lambda^* \mathcal{M}u^*, \quad (2.76)$$

where $\mathcal{A}^*$ denotes the Hilbert adjoint of $\mathcal{A}$. In the present setting, however, the auxiliary eigenproblems we consider in Algorithm 2 are symmetric. When considering symmetric eigenproblems, the primal and dual eigenproblems coincide. This leads to the following symmetric version of the DWR error estimate, which is stated and proved in [5, 19]:

⁴Let X and Y be normed spaces. The linear operator T is compact if and only if it maps every bounded sequence $\{x_n\}_{n \in \mathbb{N}}$ in X onto a sequence $\{Tx_n\}_{n \in \mathbb{N}}$ in Y that has a convergent subsequence. The embedding $V \hookrightarrow \mathcal{H}$ is compact when the embedding map from V to $\mathcal{H}$ is compact.

Corollary 2.4.1 (DWR Method for a Symmetric Eigenproblem [5, 19]). *Consider an eigenpair $\{u_h, \lambda_h\} \in V_h \times \mathbb{R}$ that solves (2.75). Define the primal eigenvalue residual as*

$$\rho(u_h, \lambda_h)(\cdot) := a(u_h, \cdot) - \lambda_h m(u_h, \cdot). \quad (2.77)$$

We have the error representation

$$(\lambda - \lambda_h)(1 - \xi_h) = \rho(u_h, \lambda_h)(u - \varphi_h), \quad (2.78)$$

for an arbitrary $\varphi_h \in V_h$, with the remainder term

$$\xi_h = \frac{1}{2} m(u - u_h, u - u_h). \quad (2.79)$$

With this result, we can estimate the error in approximating an eigenvalue of (2.73) with an eigenvalue of (2.75). Furthermore, we can localize the global error estimate (2.78) to obtain local error contributions for each cell of the mesh that underlies the current discretization. With this in mind, the localization of (2.78) can be used to drive an adaptive mesh refinement technique.

2.4.3 Localization of Error Estimates

A common approach for localizing DWR error estimates is to decompose the integrals appearing in the error estimate over the cells of the mesh and to integrate by parts on each cell. This results in a sum of residual contributions from both the cells and the facets of the underlying mesh. For worked examples of this localization procedure, the curious reader may consult [5, 19, 25, 14].

Traditionally, this localization procedure had to be performed by hand on a problem-by-problem basis. Consequently, the local error indicators required for adaptive mesh refinement had to be re-derived and implemented for each new problem of interest. This difficulty was addressed by Marie Rognes and Anders Logg, who devised a method for automating the localization of DWR error estimates [25].

Let $\mathcal{T}_h$ denote the mesh underlying the current discretization, let T index each cell of the mesh, and let ∂T denote the boundary of T . In [25], Rognes and Logg noticed that for a large class of problems, the localization procedure described above takes a specific form. In particular, they noticed that localization results in an error estimate that can be written as a sum of contributions from the cells and the facets of the mesh. In the present setting, this observation yields the following localization of (2.78):

$$\rho(u_h, \lambda_h)(u - \varphi_h) = \sum_{T \in \mathcal{T}_h} \{ (R(u_h, \lambda_h), u - \varphi_h)_{L^2(T)} + (r(u_h, \lambda_h), u - \varphi_h)_{L^2(\partial T)} \}, \quad (2.80)$$

where $R(u_h, \lambda_h)$ and $r(u_h, \lambda_h)$ denote the localized cell and facet residuals, respectively [14]. Here, $(\cdot, \cdot)_{L^2(T)}$ and $(\cdot, \cdot)_{L^2(\partial T)}$ denote the L^2 inner product, where the integration is performed over T and ∂T , respectively.

With this observation, Rognes and Logg show that by solving local problems on each cell T of the mesh, we can compute piecewise polynomial approximations R_T and $r_{\partial T}$ to $R(u_h, \lambda_h)|_T$ and $r(u_h, \lambda_h)|_{\partial T}$, respectively [25]. In doing so, the primal residual can be approximated via

$$\rho(u_h, \lambda_h)(u - \varphi_h) \approx \sum_{T \in \mathcal{T}_h} \{(R_T, u - \varphi_h)_{L^2(T)} + (r_{\partial T}, u - \varphi_h)_{L^2(\partial T)}\}, \quad (2.81)$$

which yields the approximate local error indicators

$$\eta_T := |(R_T, u - \varphi_h)_{L^2(T)} + (r_{\partial T}, u - \varphi_h)_{L^2(\partial T)}|. \quad (2.82)$$

These local error indicators can be used to indicate the regions of the mesh that contribute the most to the error in the desired quantity of interest. As a result, they can be used to drive an adaptive mesh refinement technique based on the error in the desired quantity of interest.

Rognes and Logg's automated localization procedure was implemented in Firedrake as a part of Joseph Flood's Master's Thesis [14]. This implementation has since been further developed by Patrick Farrell and Pablo Brubeck [13]. For this project, we build upon the existing implementation and adapt it to the auxiliary eigenproblems arising in Algorithm 2. We therefore omit the details of the local problems used to obtain (2.82) and refer the reader to [14, 25, 13] for further details on the localization procedure and its implementation.

2.4.4 Approximating the Exact Errors

Notice that by applying Galerkin orthogonality, the primal residual in (2.77) can be written as $\rho(u_h, \lambda_h)(u - \varphi_h) = \rho(u_h, \lambda_h)(u)$ for any $\varphi_h \in V_h$. Thus, we expect choice of φ_h to have very little impact on the global error estimate (2.78). However, we do not expect Galerkin orthogonality to hold on each cell of the mesh. Thus, we expect the choice of φ_h to impact the local error estimates (2.82) and the adaptive refinement procedure that results.

Two common choices for φ_h are $\varphi_h := u_h$ and $\varphi_h := I_h u$, where $I_h u$ denotes the interpolation of u onto V_h . These choices give the weights $u - u_h$ and $u - I_h u$ in (2.78) and (2.82), respectively. The first represents the discretization error in the eigenfunction, while the second represents the component of u that is not represented

by its interpolant in V_h . In [24], it is shown that choosing $\varphi_h := u_h$ allows for the separation of the error incurred by discretization and the error incurred from inexactly solving (2.75). As a result, we adopt the first choice and take $u - \varphi_h = u - u_h$ in our error estimates⁵.

To compute either $u - u_h$ or $u - I_h u$, we need the eigenfunction u of the infinite-dimensional problem (2.73). In practice, we do not have access to u . If we did, there would be no need to approximate it. Thus, for the practical computation of (2.78) and (2.82), we have to approximate the terms involving u . In particular, we have to approximate $e := u - u_h$ in the remainder term of (2.78), as well as $e_h := u - \varphi_h$ in (2.78) and (2.82).

There are many strategies for approximating e and e_h , each of which is discussed in [5, 19]. The most common approaches approximate u in an ‘enriched’ finite element space, V_h^+ , that is obtained by increasing the degree of the piecewise polynomial basis used to construct V_h . To obtain an ‘enriched’ approximation u_h^+ to u in V_h^+ , there are two common approaches. We discuss both in turn:

1. **Global Higher-Order Approximation.** The simplest approach is to solve for u_h^+ by solving (2.75) with the choice of V_h^+ instead of V_h . An advantage of this approach is that it is easy to implement in Firedrake, which takes care of the finite element assembly, solvers, and interpolation required for $\varphi_h = I_h u_h^+$. A disadvantage is the cost of re-solving (2.75) in a more expensive space.
2. **Local Higher-Order Approximation.** An alternative approach is to use patch-wise extrapolation to approximate u_h^+ using u_h , without having to re-solve (2.75). Rognes and Logg describe a least-squares implementation of this approach in [25].

Following Flood, Farrell, and Brubeck, we choose the first of the two approaches because of its straightforward implementation in Firedrake [14, 13]. The patch-wise extrapolation required for the second choice is not yet readily available in Firedrake and could thus be a topic of future work.

With the choices $\varphi_h = u_h$ and $u \approx u_h^+$, we obtain the following computable DWR eigenvalue error estimate for the eigenvalue error in (2.78):

$$\eta := \frac{\rho(u_h, \lambda_h)(u_h^+ - u_h)}{1 - \frac{1}{2}m(u_h^+ - u_h, u_h^+ - u_h)}. \quad (2.83)$$

We use (2.83) and its localization to drive an adaptive mesh refinement technique. We develop this technique next.

⁵We note that both choices were tested numerically and produced similar results.

Chapter 3

Methodology

In this chapter, we develop our methodology for pairing our adaptive mesh refinement technique of Section 3.1.3 with an outer refinement technique for selecting the test points used in Algorithm 2. In the first section, we couple Algorithm 2 with our adaptive mesh refinement technique. In the second section, we combine this methodology with an outer refinement procedure. In doing so, we develop an algorithm for computing pseudospectra both accurately and efficiently.

3.1 An Adaptive Refinement Technique for Colbrook's Algorithm

As discussed, Algorithm 2 approximates the pseudospectra of an operator without suffering from the pitfalls of the discretize-then-solve approach. To obtain these approximations, we compute the smallest eigenvalues of auxiliary eigenproblems for the two folded operators $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$. However, as shown in Section 2.3.4, if the discretization of these eigenproblems is too coarse, Algorithm 2 may produce poor pseudospectral approximations.

To improve these approximations, we apply the DWR method to Algorithm 2. To do so, we recall that for a fixed z in Algorithm 2, we compute the smallest eigenvalues, $\mu_h^{(1)}(z)$ and $\mu_h^{(2)}(z)$, of the auxiliary eigenproblems, (2.62) and (2.64), respectively. By identifying $a(\cdot, \cdot)$ in Section 2.4 with the sesquilinear formulations of $Q_{z,A}^{(1)}(\cdot, \cdot)$ and $Q_{z,A}^{(2)}(\cdot, \cdot)$ in Algorithm 2, we can apply the discussion of Section 2.4 to the auxiliary eigenproblems in Algorithm 2. In doing so, we can compute DWR error estimates for $\mu_h^{(1)}(z)$ and $\mu_h^{(2)}(z)$.

We use these error estimates in two ways. First, we use the error estimates to correct the computed eigenvalues, thereby improving the computed eigenvalues, and thus the values of $\Phi_h(z, A)$ in Algorithm 2. Second, we use localized error estimates to drive an adaptive mesh refinement technique that concentrates the computational effort in regions of the mesh where the computed eigenvalue error is large.

In this section, we outline the details of our application of the DWR method to Algorithm 2. In the first subsection, we explore the needed assumptions for the successful application of the DWR error estimate in (2.78) to the auxiliary eigenproblems in Algorithm 2. In the second subsection, we discuss our adaptation of (2.78) to account for eigenvalues with multiplicity greater than one. Finally, in the last

three subsections, we outline our methodology for pairing DWR with Algorithm 2 and demonstrate its success when applied to the Maxwell eigenproblem.

3.1.1 The Needed Assumptions

The DWR error estimate in (2.78) is formulated for infinite-dimensional eigenproblems as opposed to infinite-dimensional spectral computations. In particular, (2.78) is derived under the assumption that the desired eigenvalue, λ , has an associated eigenfunction u . As a result, the error estimate requires an approximation of the eigenvalue, λ_h , and an approximation of the corresponding discrete eigenfunction, u_h .

In Algorithm 2, the approximate bounds on the distance to the spectrum, $\Phi_h(z, A)$, are obtained by approximating the bottom of the spectrum of the two folded operators $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$. There is no requirement that these spectral bottoms be eigenvalues of finite multiplicity with associated eigenfunctions. For example, there can be values of z in the essential spectrum of an operator A for which $\|(A - zI)x_n\|_{\mathcal{H}} \rightarrow 0$, $\|x_n\|_{\mathcal{H}} = 1$, and $\{x_n\}_{n \in \mathbb{N}} \subset D(A)$ converges weakly to zero¹ [9].

With this in mind, it is not clear how to apply the DWR error estimate for eigenvalues when the computed eigenfunction u_h does not reflect a genuine eigenfunction u . We therefore restrict our attention to operators for which the bottom of the spectrum of each corresponding folded operator is an isolated eigenvalue with finite multiplicity. The precise conditions on the operator A for which this is the case are beyond the scope of the present work, and are left as a task for future work. Furthermore, in future work, we aim to extend the current DWR framework to account for folded operators whose spectral bottom is an element of the essential spectrum.

3.1.2 The DWR Method for Multiple Eigenvalues

Even if we assume that the smallest eigenvalues of the auxiliary eigenproblems in Algorithm 2 approximate finite-multiplicity eigenvalues of the corresponding infinite-dimensional folded operators, we still need to account for the possibility that the target eigenvalue has multiplicity greater than one.

To see this, consider the case in which the closed and densely defined linear operator $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$ is self-adjoint and $z \in \mathbb{R}$. The folded operator for a given $z \in \mathbb{R}$ is $(A - zI)^*(A - zI) = (A - zI)(A - zI)^* = (A - zI)^2$. Thus, if λ is an eigenvalue of A with multiplicity greater than one, then $(\lambda - z)^2$ is an eigenvalue of

¹This phenomenon is called a *Weyl sequence* and can be used to characterize the essential spectrum of an operator. For more information, see Chapter 8 of [9].

the folded operator with multiplicity that is at least the multiplicity of λ .

Let λ be the smallest eigenvalue of an infinite-dimensional eigenproblem. If λ has a multiplicity $m > 1$, then a finite element discretization yields m eigenvalues, $\lambda_{h,i}$ for $i = 1, 2, \dots, m$, that approximate λ , each with a corresponding discrete eigenfunction, $u_{h,i}$ [7, 18]. Since the eigenspace associated with λ is m -dimensional, each $u_{h,i}$ may approximate any linear combination of eigenfunctions that belong to the target eigenspace.

To use DWR to estimate the error in the computed eigenvalue, we require an enriched approximation u_h^+ that corresponds to the chosen discrete eigenfunction u_h . When λ is an eigenvalue with multiplicity one, this matching is simple: we compute u_h and u_h^+ , normalize them, and align their phases. When λ has multiplicity $m > 1$, the primal and enriched solves may approximate different bases for the same m -dimensional eigenspace. As a result, it is more challenging to choose a u_h^+ that behaves as an enriched approximation of the chosen u_h .

To overcome this challenge, we propose a least squares approach to matching u_h with an enriched approximation u_h^+ . In particular, given the computed enriched basis $E_h^+ := \{u_{h,1}^+, u_{h,2}^+, \dots, u_{h,m}^+\}$, we take u_h^+ to be the solution of the following problem:

$$\arg \min_{v \in \text{span}(E_h^+)} \|u_h - v\|_{\mathcal{H}}, \quad (3.1)$$

where $\|\cdot\|_{\mathcal{H}}$ is the norm associated with the underlying Hilbert space $\mathcal{H}$.

To solve this problem, we formulate it as a matrix problem. For u_h^+ to solve (3.1), the residual $r^+ := u_h - u_h^+$ must be orthogonal to every direction in $\text{span}(E_h^+)$. This follows from the fact that if u_h^+ is the best approximation of u_h in $\text{span}(E_h^+)$, then moving in any direction in $\text{span}(E_h^+)$ cannot decrease the residual. Thus, we require $(r^+, u_{h,i}^+)_{\mathcal{H}} = 0$ for $i = 1, 2, \dots, m$. Substituting in $r^+ = u_h - \sum_{j=1}^m \alpha_j u_{h,j}^+$, we obtain the m -dimensional matrix problem

$$\sum_{j=1}^m \alpha_j (u_{h,j}^+, u_{h,i}^+)_{\mathcal{H}} = (u_h, u_{h,i}^+)_{\mathcal{H}}, \quad i = 1, 2, \dots, m, \quad (3.2)$$

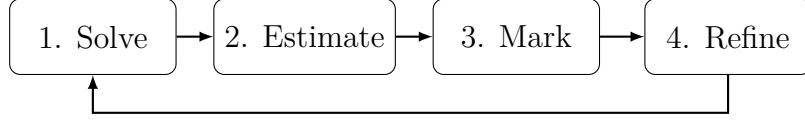
which we solve for the coefficients $\alpha_1, \alpha_2, \dots, \alpha_m$, that define u_h^+ .

In solving (3.2), we obtain an enriched approximation u_h^+ that represents the chosen primal eigenfunction u_h within the computed enriched eigenbasis. This allows us to apply the DWR error estimate (2.83) to eigenvalues with multiplicity greater than one. As a result, (2.83) can be used to estimate the error in the smallest eigenvalues arising in Algorithm 2. The details of this implementation are discussed in the next subsection.

3.1.3 An Adaptive Mesh Refinement Technique

With our DWR error estimate and its localization at hand, we can estimate the error in the computed smallest eigenvalues of the auxiliary eigenproblems arising in Algorithm 2. In doing so, we can use the error estimate to correct the computed eigenvalues and to drive an adaptive refinement technique at each fixed value of z in Algorithm 2.

Standard adaptive mesh refinement techniques follow an adaptive feedback loop:



Let $j \in \{1, 2\}$ index the j -th auxiliary eigenproblem for a set $z \in \text{Grid}$ in Algorithm 2. For each j , the four steps of our adaptive mesh refinement technique take the following form:

1. **Solve:** For a fixed $z \in \text{Grid}$ in Algorithm 2, solve the j -th auxiliary eigenproblem on its current finite element space $V_h^{(j)}$. Retain the smallest eigenvalue $\mu_h^{(j)}(z)$ and the corresponding discrete eigenfunction $u_h^{(j)}$. Construct the enriched space $V_h^{(j),+}$ by increasing the polynomial degree that defines $V_h^{(j)}$. Solve the j -th auxiliary eigenproblem in the enriched space.
2. **Estimate:** Solve the least-squares problem (3.1) to compute the enriched approximation that best matches $u_h^{(j)}(z)$. Apply the DWR error estimate (2.83) to obtain the following global error estimate:

$$\eta^{(j)}(z) \approx \mu^{(j)}(z) - \mu_h^{(j)}(z). \quad (3.3)$$

Using this error estimate, compute the corrected eigenvalue, $\mu_{h,c}^{(j)}(z)$, as follows:

$$\mu_{h,c}^{(j)}(z) := \eta^{(j)}(z) + \mu_h^{(j)}(z), \quad (3.4)$$

where $\mu^{(j)}(z)$ denotes the target eigenvalue. Then, use the method of Rognes and Logg to compute the local error indicators $\eta_T^{(j)}(z)$, as given by (2.82), for each cell T in the mesh that defines $V_h^{(j)}$ [25, 14, 13].

3. **Mark:**² Mark the cells that have a local error that is considered too large. For our implementation, we mark the cells whose local error estimate satisfies

$$\eta_T^{(j)}(z) \geq \theta \max_k \eta_k^{(j)}(z), \text{ for some } \theta \in [0, 1]. \quad (3.5)$$

²This marking scheme was obtained from [12] where it is said to be simpler than the famous Dörfler marking routine, but performs similarly in practice [10, 12].

Unless otherwise stated, we take $\theta = 1/2$.

4. **Refine:** Refine the marked cells, thereby defining a new finite element function space for the j -th auxiliary eigenproblem. To refine our marked cells, we employ Netgen's refinement technique, as implemented by Flood, Farrell, and Brubeck [23, 27, 13].

To implement this procedure, we iterate between Steps 1 $\rightarrow$ 4 until a user-defined stopping criterion has been satisfied. In the next subsection, we demonstrate the success of this methodology by applying it to the Maxwell eigenproblem.

3.1.4 Application to the Maxwell Eigenproblem

We now return to the Maxwell eigenproblem stated in (2.18). In Section 2.3.4, we applied Algorithm 2 to the Maxwell operator (2.66). The numerical experiments in that section showed that Colbrook's approach avoids the spectral pollution exhibited by the discretize-then-solve approach of Section 2.2.1. However, as shown in Figure 2.2, coarse discretizations of the auxiliary eigenproblems can produce poor bounds on the distance to the spectrum. In an effort to compute these bounds accurately, without resorting to extensive uniform refinement, we couple Algorithm 2 with the adaptive mesh refinement technique in Section 3.1.3.

To do so, we take the starting mesh to be an unstructured triangular mesh generated by Netgen, with a maximum element diameter of $\pi/32$ [27]. We take V_h to be the space of linear Lagrange elements satisfying the zero tangential trace boundary condition in (2.18). The enriched space V_h^+ is obtained by increasing the polynomial degree of V_h by one. As in Section 2.3.4, we take $\text{Grid} \subset \mathbb{R}$. In particular, we set

$$\text{Grid} = \{2 + jh : j = 0, \dots, 100\}, \quad h = 0.02. \quad (3.6)$$

Notice that, as opposed to the choice of Grid in Section 2.3.4, i.e., (2.71), we no longer consider test points near $z = 0$. Recall from Section 3.1.1 that our adaptive mesh refinement technique can only be applied when the bottom of the spectrum of the folded operator is an eigenvalue with finite multiplicity. For the Maxwell operator (2.66), the choice of Grid in (3.6) avoids the case in which this smallest eigenvalue has infinite multiplicity. Further details on this choice are given in Appendix B.

To couple Algorithm 2 with the adaptive mesh refinement technique of Section 3.1.3, we iterate over each $z \in \text{Grid}$. At each z , we compute the smallest eigenvalues, $\mu_h(z)$ and $\mu_h^+(z)$, of the auxiliary eigenproblem, (2.69), posed on V_h and V_h^+ , respectively. To do so, we write (2.69) in UFL and ask Firedrake to solve the resulting generalized eigenproblem using SLEPc's Krylov-Schur method with a shift-and-invert

spectral transformation [17, 18]. We then solve the least squares problem (3.1) to match the computed eigenfunction $u_h \in V_h$ with the enriched $u_h^+ \in V_h^+$.

Using u_h and u_h^+ , we compute the DWR error estimate $\eta(z) \approx \mu(z) - \mu_h(z)$, where $\mu(z)$ denotes the target eigenvalue. Using this error estimate, we compute the corrected eigenvalue $\mu_{h,c}(z) := \eta(z) + \mu_h(z)$. Then we compute the corrected approximation $\Phi_{h,c}(z, A) = \sqrt{\mu_{h,c}(z)}$.

Using the localization procedure of Rognes and Logg, we compute a local error indicator $\eta_T(z)$ for each cell T of the mesh that defines V_h and V_h^+ . We then mark and refine all cells for which (3.5) holds. This results in a new mesh and, consequently, a new V_h and V_h^+ for the discretization of the auxiliary eigenproblem.

This process is repeated for each $z \in \text{Grid}$ until a user-defined stopping criterion has been satisfied. In the next chapter, we develop a specific stopping criterion that couples the accuracy required for the auxiliary eigenproblems with the desired resolution of the ϵ -pseudospectrum. In the present setting, whose aim is to demonstrate the benefit of pairing Algorithm 2 with adaptive refinement of the auxiliary eigenproblems, we perform five steps of the adaptive mesh refinement technique at each $z \in \text{Grid}$. The results of this experiment are displayed in Figure 3.1. Notice that the DWR-corrected bound is substantially more accurate than the bound obtained on a uniform mesh. This result further motivates the coupling of Colbrook's algorithm with the adaptive mesh refinement technique of Section (3.1.3).

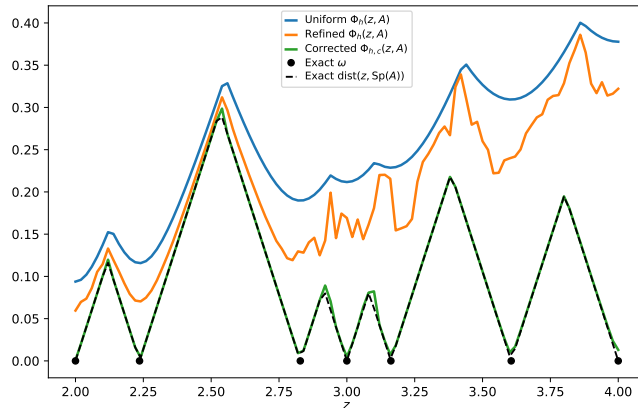


Figure 3.1: The results from applying five iterations of our adaptive mesh refinement technique of Section 3.1.3 to Algorithm 2 for the Maxwell operator (2.66). The blue curve shows $\Phi_h(z, A)$ when no adaptivity is applied. The orange curve shows $\Phi_h(z, A)$ after five iterations of adaptive mesh refinement. The green curve shows the corresponding corrected approximation $\Phi_{h,c}(z, A)$. The black dashed curve shows the exact distance $\text{dist}(z, \text{Sp}(A))$ and the black dots indicate the exact eigenvalues.

3.2 Coupling Inner and Outer Adaptivity for Colbrook's Algorithm

With the adaptive mesh refinement technique of Section 3.1.3 at hand, we can efficiently solve the auxiliary eigenproblems that arise in Algorithm 2. However, two questions remain. First, how should we pick the test points in the complex plane so that the computational effort of Algorithm 2 is concentrated near the desired ϵ -pseudospectrum? Second, for a given test point z , to what accuracy should we solve the corresponding auxiliary eigenproblems? For example, if z is far from the ϵ -pseudospectrum, there is little benefit in considering nearby points in the complex plane. Furthermore, for a far-away z , there is little benefit in computing an accurate approximation of $\gamma(z, A)$.

In this section, we answer these two questions. Our answer to the first question will determine a method for adaptively picking test points in the complex plane. Our answer to the second question will determine the relationship between ‘outer refinement’ of the complex plane and ‘inner refinement’ of the auxiliary eigenproblems that arise at each test point. Together, these ideas give rise to our proposed methodology, which combines both inner and outer refinement with Algorithm 2 to efficiently resolve the pseudospectra of infinite-dimensional operators.

3.2.1 The Inner Refinement Criteria

Our adaptive mesh refinement technique in Section 3.1.3 estimates the error in the computed smallest eigenvalues of the auxiliary eigenproblems arising in Algorithm 2. In particular, for each auxiliary eigenproblem, (2.62) and (2.64), we compute

$$\eta^{(1)}(z) \approx \mu^{(1)}(z) - \mu_h^{(1)}(z), \text{ and } \eta^{(2)}(z) \approx \mu^{(2)}(z) - \mu_h^{(2)}(z), \quad (3.7)$$

respectively. In our adaptive mesh refinement technique, we use (3.7) to correct our computed eigenvalues as follows:

$$\mu_{h,c}^{(1)}(z) := \eta^{(1)}(z) + \mu_h^{(1)}(z), \text{ and } \mu_{h,c}^{(2)}(z) := \eta^{(2)}(z) + \mu_h^{(2)}(z). \quad (3.8)$$

This technique makes use of error estimates for the computed eigenvalues of the auxiliary eigenproblems. However, the quantity of interest in Algorithm 2 is $\Phi_h(z, A)$ in (2.65). As a result, we want to use (3.7) to approximate the error in $\Phi_h(z, A)$, that is,

$$|\gamma(z, A) - \Phi_h(z, A)| = \left| \sqrt{\min\{\mu^{(1)}(z), \mu^{(2)}(z)\}} - \sqrt{\min\{\mu_h^{(1)}(z), \mu_h^{(2)}(z)\}} \right|. \quad (3.9)$$

Furthermore, we want to use the error in $\Phi_h(z, A)$ to drive the adaptive mesh refinement technique of Section 3.1.3.

To use (3.7) to construct an error bound for (3.9), we first note that for scalars $a, b, c \in \mathbb{R}$, $|a - b| \leq c$ implies that $|\sqrt{a} - \sqrt{b}| \leq \sqrt{c}$. Thus, to construct an error bound for (3.9), we can construct an error bound for

$$|\gamma^2(z, A) - \Phi_h^2(z, A)| = \left| \min\{\mu^{(1)}(z), \mu^{(2)}(z)\} - \min\{\mu_h^{(1)}(z), \mu_h^{(2)}(z)\} \right|. \quad (3.10)$$

To do so, we recall that for any $a, b \in \mathbb{R}$, $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$. Thus, we can write (3.10) as

$$|\gamma^2(z, A) - \Phi_h^2(z, A)| = \left| \frac{\mu^{(1)}(z) + \mu^{(2)}(z) - |\mu^{(1)}(z) - \mu^{(2)}(z)|}{2} - \frac{\mu_h^{(1)}(z) + \mu_h^{(2)}(z) - |\mu_h^{(1)}(z) - \mu_h^{(2)}(z)|}{2} \right|. \quad (3.11)$$

Rearranging and applying the triangle inequality yields

$$|\gamma^2(z, A) - \Phi_h^2(z, A)| \leq \frac{1}{2} \left(\left| \mu^{(1)}(z) - \mu_h^{(1)}(z) \right| + \left| \mu^{(2)}(z) - \mu_h^{(2)}(z) \right| + \left| \left| \mu^{(1)}(z) - \mu^{(2)}(z) \right| - \left| \mu_h^{(1)}(z) - \mu_h^{(2)}(z) \right| \right|. \quad (3.12)$$

Then, applying the reverse triangle inequality, followed by triangle inequality, we obtain

$$|\gamma^2(z, A) - \Phi_h^2(z, A)| \leq \left| \mu^{(1)}(z) - \mu_h^{(1)}(z) \right| + \left| \mu^{(2)}(z) - \mu_h^{(2)}(z) \right|. \quad (3.13)$$

Thus, our desired bound for (3.9) is

$$|\gamma(z, A) - \Phi_h(z, A)| \leq \sqrt{\left| \mu^{(1)}(z) - \mu_h^{(1)}(z) \right| + \left| \mu^{(2)}(z) - \mu_h^{(2)}(z) \right|}. \quad (3.14)$$

Notice that (3.14) is not computable because we do not know $\mu^{(1)}(z)$ and $\mu^{(2)}(z)$. To obtain a practical, but heuristic, bound, we use (3.7) to write

$$\begin{aligned} |\gamma(z, A) - \Phi_h(z, A)| &\leq \sqrt{\left| \mu^{(1)}(z) - \mu_h^{(1)}(z) \right| + \left| \mu^{(2)}(z) - \mu_h^{(2)}(z) \right|} \\ &\approx \sqrt{|\eta^{(1)}(z)| + |\eta^{(2)}(z)|}. \end{aligned} \quad (3.15)$$

Using (3.15), we can control the error in $\Phi_h(z, A)$ by controlling the error estimates $\eta^{(1)}$ and $\eta^{(2)}$ for the smallest eigenvalues of the auxiliary eigenproblems. We can therefore formulate the stopping criterion for the adaptive mesh refinement technique

of Section 3.1.3 in a way that reflects the desired accuracy for $\Phi_h(z, A)$. To do so, we must decide what accuracy we require at each test point z in the complex plane. We must also decide how refinement of the auxiliary eigenproblems should be coupled with refinement of $\text{Grid} \subset \mathbb{C}$, so that the computational effort of Algorithm 2 is concentrated near the ϵ -pseudospectrum of interest. We discuss these decisions next.

3.2.2 The Outer Refinement Criteria

Given a fixed $\epsilon > 0$, we want to use Algorithm 2 to compute the ϵ -pseudospectrum of a closed and densely defined linear operator A . To do so, we propose an approach that couples refinement of $\text{Grid} \subset \mathbb{C}$ with our adaptive mesh refinement technique of Section 3.1.3. In this section, we outline the specifics of this coupling.

We begin by fixing a bounded region $\mathcal{D}$ in the complex plane where we want to resolve the ϵ -pseudospectrum of A . Let $\mathcal{T}_k$ denote a triangulation of $\mathcal{D}$ at the outer refinement level k . For each triangle $K \in \mathcal{T}_k$, let z_K denote its barycentre and define its diameter by

$$h_K := \max_{z, w \in K} |z - w|. \quad (3.16)$$

To apply Algorithm 2 at the outer refinement level k , we take

$$\text{Grid} = \{z_K : K \in \mathcal{T}_k\}. \quad (3.17)$$

Thus, for each triangle $K \in \mathcal{T}_k$, we consider the barycentre z_K and compute $\Phi_h(z_K, A)$. This setup is illustrated in Figure 3.2.

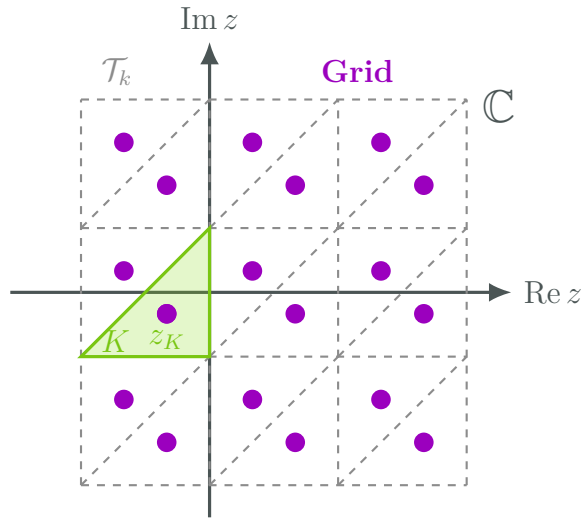


Figure 3.2: A representative triangulation of the complex plane.

Consider a triangle $K \in \mathcal{T}_k$ with barycentre z_K , as illustrated in Figure 3.2. We want to determine whether the ϵ -pseudospectral contour, $\gamma(z, A) = \epsilon$, passes through this triangle. To do so, we first make use of the following result:

Lemma 3.2.1 ($\gamma(\cdot, A)$ is 1-Lipschitz³). *Let $\gamma(\cdot, A)$ be defined as in (2.37). Then $\gamma(\cdot, A)$ is 1-Lipschitz, that is,*

$$|\gamma(z, A) - \gamma(w, A)| \leq |z - w|, \quad \forall z, w \in \mathbb{C}. \quad (3.18)$$

Proof. Recall from (2.37) that $\gamma(z, A) = \min \left\{ \sqrt{\mu^{(1)}(z)}, \sqrt{\mu^{(2)}(z)} \right\}$, where

$$\mu^{(1)}(z) = \inf_{u \in D(A), \|u\|_{\mathcal{H}}=1} \|(A - zI)u\|_{\mathcal{H}}^2, \quad (3.19)$$

$$\mu^{(2)}(z) = \inf_{u \in D(A^*), \|u\|_{\mathcal{H}}=1} \|(A - zI)^*u\|_{\mathcal{H}}^2. \quad (3.20)$$

Notice that

$$(A - zI)u = (A - wI)u + (w - z)u. \quad (3.21)$$

Thus,

$$\|(A - zI)u\|_{\mathcal{H}} \leq \|(A - wI)u\|_{\mathcal{H}} + \|(w - z)u\|_{\mathcal{H}}. \quad (3.22)$$

Taking the infimum over all $u \in D(A)$ such that $\|u\|_{\mathcal{H}} = 1$, we obtain

$$\sqrt{\mu^{(1)}(z)} \leq \sqrt{\mu^{(1)}(w)} + |z - w|. \quad (3.23)$$

By interchanging z and w , we find that $\sqrt{\mu^{(1)}(\cdot)}$ is 1-Lipschitz:

$$\left| \sqrt{\mu^{(1)}(z)} - \sqrt{\mu^{(1)}(w)} \right| \leq |z - w|. \quad (3.24)$$

Similar arguments show that $\sqrt{\mu^{(2)}(z)}$ is 1-Lipschitz. Since the minimum of two 1-Lipschitz functions is itself 1-Lipschitz, we have our desired result. $\square$

Assume that the pseudospectral contour $\gamma(z, A) = \epsilon$ does pass through triangle K . That is, assume that there is a $z_{\epsilon} \in K$ such that $\gamma(z_{\epsilon}, A) = \epsilon$. Using Lemma 3.2.1, we have that

$$|\Phi_h(z_K, A) - \epsilon| \leq |\Phi_h(z_K, A) - \gamma(z_K, A)| + |\gamma(z_K, A) - \epsilon| \quad (3.25)$$

$$= |\Phi_h(z_K, A) - \gamma(z_K, A)| + |\gamma(z_K, A) - \gamma(z_{\epsilon}, A)| \quad (3.26)$$

$$\leq |\Phi_h(z_K, A) - \gamma(z_K, A)| + |z_K - z_{\epsilon}| \quad (3.27)$$

$$\leq |\Phi_h(z_K, A) - \gamma(z_K, A)| + h_K, \quad (3.28)$$

³This result is stated as a part of Exercise 1.8 in [9]. We developed the proof using arguments similar to the proof of Lemma 3.4 in [28].

where h_K is as defined in (3.16). Using (3.15), we write

$$|\Phi_h(z_K, A) - \epsilon| \lesssim \sqrt{|\eta^{(1)}(z_K)| + |\eta^{(2)}(z_K)|} + h_K, \quad (3.29)$$

where we have used ‘ $\lesssim$ ’ to indicate that (3.15) is a heuristic error bound. With this, we can treat (3.29) as a necessary condition for the pseudospectral contour $\gamma(z, A) = \epsilon$ to pass through triangle K .

Next, consider an arbitrary $w \in K$. Using (3.29) and (3.15), we have that

$$|\gamma(w, A) - \Phi_h(z_K, A)| \leq |\gamma(w, A) - \gamma(z_K, A)| + |\gamma(z_K, A) - \Phi_h(z_K, A)| \quad (3.30)$$

$$\leq |w - z_K| + |\gamma(z_K, A) - \Phi_h(z_K, A)| \quad (3.31)$$

$$\leq h_K + |\gamma(z_K, A) - \Phi_h(z_K, A)| \quad (3.32)$$

$$\lesssim h_K + \sqrt{|\eta^{(1)}(z_K)| + |\eta^{(2)}(z_K)|} := R. \quad (3.33)$$

Rearranging, we obtain

$$\Phi_h(z_K, A) - R \leq \gamma(w, A) \leq \Phi_h(z_K, A) + R. \quad (3.34)$$

From this result, we obtain three cases that we can use to determine whether K can be classified as lying inside or outside the ϵ -pseudospectrum, or whether K should be refined. We discuss each case in turn:

1. Triangle K lies entirely within the ϵ -pseudospectrum. This occurs when

$$\Phi_h(z_K, A) + R < \epsilon, \quad (3.35)$$

which indicates that $\gamma(w, A) < \epsilon$ for all $w \in K$.

2. Triangle K lies entirely outside the ϵ -pseudospectrum. This occurs when

$$\Phi_h(z_K, A) - R > \epsilon, \quad (3.36)$$

which indicates that $\gamma(w, A) > \epsilon$ for all $w \in K$.

3. Triangle K should be refined. When the first two cases do not hold, we have that

$$|\Phi_h(z_K, A) - \epsilon| \leq R. \quad (3.37)$$

Hence, we do not have enough information to decide whether the pseudospectral contour $\gamma(z, A) = \epsilon$ passes through K . In this case, we refine K and re-check the three cases for each sub-triangle.

In the next subsection, we construct a coupling between the adaptive mesh refinement technique of Section 3.1.3 and the outer refinement strategy developed in this section. In doing so, we construct an efficient method for computing pseudospectra with Algorithm 2.

3.2.3 Our Implementation

We now construct our methodology for efficient pseudospectral computations using Algorithm 2. The method couples the inner refinement technique of Section 3.1.3 with the outer refinement technique of the previous subsection. The main idea behind this coupling is to only solve the auxiliary eigenproblems of Algorithm 2 to an accuracy that agrees with the current resolution of the complex plane.

As before, let $\mathcal{T}_k$ denote the triangulation of a bounded region $\mathcal{D} \subset \mathbb{C}$ at the outer refinement level k . Let K index each triangle with barycentre z_K and diameter h_K . Furthermore, as in (3.17), set Grid to be the collection of barycentres. For each $z_K \in \text{Grid}$, we solve for the smallest eigenvalues of the two auxiliary eigenproblems that arise in Algorithm 2. To do so, we apply the adaptive mesh refinement technique of Section 3.1.3 until the error estimates, $\eta^{(1)}(z_K)$ and $\eta^{(2)}(z_K)$ from (3.7), satisfy

$$|\eta^{(1)}(z_K)| \leq h_K^2/2, \text{ and } |\eta^{(2)}(z_K)| \leq h_K^2/2, \quad (3.38)$$

or until a prescribed maximum number of degrees of freedom has been reached. These stopping criteria have been chosen so that the error estimate for $\Phi_h(z, A)$ in (3.15) is of the same order as the $\mathcal{O}(h_K)$ uncertainty in representing the triangle K with its barycentre z_K . In this way, the auxiliary eigenproblems are not solved to a greater accuracy than what is warranted by the resolution of the complex plane.

Once (3.38) has been satisfied, we compute $\Phi_h(z_K, A)$ as in (2.65). Using the DWR-corrected eigenvalues, $\mu_{h,c}^{(1)}(z_K)$ and $\mu_{h,c}^{(2)}(z_K)$ in (3.8), we also compute the corrected approximation

$$\Phi_{h,c}(z_K, A) = \sqrt{\min \left\{ \mu_{h,c}^{(1)}(z_K), \mu_{h,c}^{(2)}(z_K) \right\}}. \quad (3.39)$$

Using the classification criteria of the previous subsection, we then decide how to classify K . If (3.35) holds, we classify K as lying entirely within the ϵ -pseudospectrum. If (3.36) holds, we classify K as lying entirely outside of the ϵ -pseudospectrum. If (3.37) holds, the current information available does not allow us to determine whether K is inside or outside the ϵ -pseudospectrum. As a result, we mark K for refinement. A triangle that has been marked for refinement is refined by connecting the midpoints of its three edges, resulting in four sub-triangles.

Once we have iterated over the barycentre of each triangle in $\mathcal{T}_k$, the sub-triangles that result from refinement are used to create the updated triangulation $\mathcal{T}_{k+1}$. We then iterate over the new set of barycentres and repeat the same classification procedure. Since the sub-triangles have a smaller diameter than the triangles in $\mathcal{T}_k$, the stopping criterion (3.38) for the adaptive mesh refinement technique will be more demanding. In this way, the outer refinement of the complex plane requires increasingly greater accuracy in the inner refinement of the auxiliary eigenproblems.

This coupling of our outer and inner refinement techniques is summarized in Algorithm 3. With this approach, the computational effort of Algorithm 2 is concentrated near the pseudospectral contour $\gamma(z, A) = \epsilon$. To implement this approach, we begin with a relatively coarse triangulation of the complex plane together with coarse discretizations of the auxiliary eigenproblems that arise at each barycentre. As we near the pseudospectral contour, both the triangulation of the complex plane and the corresponding discretizations of the auxiliary eigenproblems are refined. We continue this coupled refinement process until every triangle that would otherwise be marked for refinement has its diameter below a prescribed tolerance. At that stage, the remaining triangles are taken to approximate the ϵ -pseudospectral contour. In the next Chapter, we will demonstrate the success of this approach by applying Algorithm 3 to numerous examples.

Algorithm 3 Coupled Inner and Outer Refinement for Computing Pseudospectra

Input: Initial triangulation $\mathcal{T}_0$ of $\mathcal{D} \subset \mathbb{C}$, initial finite element space $V_h^{(j)}$ for the j -th auxiliary eigenproblem ($j \in \{1, 2\}$), $\epsilon > 0$, outer tolerance $h_{\text{tol}} > 0$, maximum number of degrees of freedom for the adaptive mesh refinement technique.

```
1: Set  $k = 0$ .
2: while there are unclassified triangles in  $\mathcal{T}_k$  do
3:   for each unclassified triangle  $K \in \mathcal{T}_k$  do
4:     Compute the barycentre,  $z_K$ , and the diameter,  $h_K$ , of  $K$ .
5:     for  $j \in \{1, 2\}$  do
6:       Starting from the finite element space  $V_h$ , apply the adaptive mesh
       refinement technique of Section 3.1.3 to the  $j$ -th auxiliary eigen-
       problem at  $z_K$  until  $|\eta^{(j)}(z_K)| \leq h_K^2/2$ , or until the prescribed max-
       imum number of degrees of freedom is reached.
7:     end for
8:     Compute  $\Phi_h(z_K, A)$  as in (2.65).
9:     Compute the corrected approximation  $\Phi_{h,c}(z_K, A)$  as in (3.9).
10:    Compute  $R$  as in (3.33).
11:    if  $\Phi_h(z_K, A) + R < \epsilon$  then
12:      Classify  $K$  as INSIDE.
13:    else if  $\Phi_h(z_K, A) - R > \epsilon$  then
14:      Classify  $K$  as OUTSIDE.
15:    else if  $h_K < h_{\text{tol}}$  then
16:      Classify  $K$  as CONTOUR.
17:    else
18:      Mark  $K$  for outer refinement.
19:    end if
20:  end for
21:  Refine each marked triangle into four sub-triangles.
22:  Form  $\mathcal{T}_{k+1}$  from the newly created sub-triangles.
23:  Set  $k \leftarrow k + 1$ .
24: end while
```

Output: A classified triangulation whose IN triangles approximate the inside of the ϵ -pseudospectrum, whose OUT triangles approximate the outside of the ϵ -pseudospectrum, and whose CONTOUR triangles approximate the ϵ -pseudospectral contour $\gamma(z, A) = \epsilon$.

Chapter 4

Numerical Examples

In this chapter, we demonstrate the success of Algorithm 3 through its application to chosen examples. For our first example, we explore the pseudospectra of a one-dimensional Dirac-type operator. For our second example, we return to the Maxwell eigenproblem introduced in Section (2.2.1).

4.1 A Dirac-Type Poisson Problem

For our first example, we consider the following self-adjoint, one-dimensional Dirac-type operator posed on $\Omega = (0, \pi)$:

$$A(u, \sigma) = (-i\sigma', -iu'), \quad (4.1)$$

where $D(A) = H_0^1(\Omega) \times H^1(\Omega)$ is dense in $\mathcal{H} = L^2(\Omega) \times L^2(\Omega)$ [20].

This operator's spectrum can be characterized using its relation to the Poisson eigenproblem. To see this, consider the eigenproblem $AU = \lambda U$, where $U = (u, \sigma)^T$. This yields the following two equations:

$$-i\sigma' = \lambda u, \quad (4.2)$$

$$-iu' = \lambda \sigma. \quad (4.3)$$

For $\lambda \neq 0$, we can solve for σ in (4.3) and substitute the result into (4.2) to obtain

$$-u'' = \lambda^2 u. \quad (4.4)$$

Notice that (4.4) is the one-dimensional Poisson eigenproblem with Dirichlet boundary conditions. Thus,

$$\lambda = \pm n, \quad n = 1, 2, \dots \quad (4.5)$$

Setting $\lambda = 0$ in (4.2) and (4.3) yields $u' = 0$ and $\sigma' = 0$. Since u satisfies Dirichlet boundary conditions, $u = 0$. However, since there are no boundary conditions imposed on σ , we can take $\sigma = c$ for any constant c . Thus, $U = (0, c)^T$ is an eigenfunction of (4.1) corresponding to $\lambda = 0$.

Since (4.1) is self-adjoint and has a compact resolvent, its spectrum consists only of its eigenvalues [8]. Furthermore, since (4.1) is normal, its ϵ -pseudospectrum is the collection of ϵ -balls centred at each of the operator's eigenvalues.

For the numerical experiment of this section, we use Algorithm 3 to compute the $\epsilon = 0.2$ pseudospectrum of (4.1). To do so, we first formulate the auxiliary

eigenproblems needed for Algorithm 2. Since (4.1) is self-adjoint, we can formulate the auxiliary eigenproblems as we did for the Maxwell operator in Section 2.3.4. That is, we consider a $V_h \subset D(A)$ and solve the following eigenproblem: find $\lambda_h \in \mathbb{R}$ and nonzero $u_h \in V_h$ such that

$$Q_{z,A}(u_h, v_h) = \lambda_h m(u_h, v_h), \quad \forall v_h \in V_h, \quad (4.6)$$

where

$$Q_{z,A}(u, v) := ((A - zI)u, (A - zI)v)_{\mathcal{H}}, \quad (4.7)$$

$$m(u, v) := (u, v)_{\mathcal{H}}. \quad (4.8)$$

Taking $\mu_h(z)$ to be the smallest eigenvalue of (4.6), our approximate bound on the distance to the spectrum is

$$\Phi_h(z, A) = \sqrt{\mu_h(z)}. \quad (4.9)$$

To implement Algorithm 3, we consider the region $\mathcal{D} = [-0.5, 4.5] \times [-1, 1]$ in the complex plane. For our initial triangulation $\mathcal{T}_0$, we partition $\mathcal{D}$ into an array of 20×8 squares and subdivide each square into two triangles (see Figure 3.2).

For the initial finite element discretization of (4.6), we partition $\Omega = (0, \pi)$ into sixteen subintervals and take $V_h \subset D(A)$ to be the corresponding quadratic Lagrange finite element space. To implement the adaptive mesh refinement technique in Algorithm 3, we introduce the enriched space V_h^+ , obtained by increasing the polynomial degree of V_h by one. At each barycentre z_K of a triangle $K \in \mathcal{T}_k$, we compute the smallest eigenvalues, $\mu_h(z_K)$ and $\mu_h^+(z_K)$, of the auxiliary eigenproblem (4.6) posed in V_h and V_h^+ , respectively. To do so, we write (4.6) in UFL and ask Firedrake to solve the resulting generalized eigenproblem using SLEPc's Krylov-Schur method with a shift-and-invert spectral transformation [17, 18]. We then compute the DWR error estimate $\eta(z_K)$, whose localization drives the adaptive mesh refinement technique described in Section 3.1.3.

For the outer tolerance, which controls the diameter of the triangles in the complex plane, we take $h_{\text{tol}} = 0.01$. Since (4.1) is self-adjoint, the inner stopping tolerance (3.38) becomes

$$|\eta(z_K)| \leq h_K^2. \quad (4.10)$$

The results of this numerical experiment are displayed in Figure 4.1. The algorithm performed seven levels of outer refinement before the remaining unclassified triangles reached the diameter tolerance and were taken to approximate the pseudospectral contour. As expected, the triangles in the complex plane are refined as

we approach the desired ϵ -pseudospectral contour from both inside and outside the ϵ -pseudospectrum. Furthermore, as the triangulation of the complex plane is refined, the corresponding adaptive eigensolves at each barycentre must satisfy a stricter stopping criterion. In this way, our coupling of inner and outer refinement in Algorithm 3 focuses the computational effort of Algorithm 2 near the pseudospectral contour of interest.

Notice that in Figure 4.1 there is a slight deterioration in the resolution of the pseudospectral contour as $\text{Re}(z)$ increases. As n increases, the eigenfunctions corresponding to the eigenvalues (4.5) become increasingly oscillatory. We therefore suspect that the auxiliary eigensolves, and thus the pseudospectral computations, become more demanding near these eigenvalues. We will see this behaviour again in applying Algorithm 3 to the Maxwell eigenproblem.

4.2 The Maxwell Eigenproblem

We now return to the Maxwell eigenproblem introduced in Section 2.2.1. Recall that (2.66) is normal. Thus, its ϵ -pseudospectrum is the union of ϵ -balls centred at each spectral value of the operator.

For our current numerical experiment, we use Algorithm 3 to compute the $\epsilon = 0.2$ pseudospectrum of the Maxwell operator (2.66). To do so, we consider the region $\mathcal{D} = [2, 4] \times [-1, 1]$ in the complex plane. For our initial triangulation $\mathcal{T}_0$, we partition $\mathcal{D}$ into a 10×10 array of squares and subdivide each square into two triangles (see Figure 3.2). Furthermore, we set $h_{\text{tol}} = 0.02$ for the outer tolerance.

The formulation and implementation of the auxiliary eigenproblem (2.69) is as in Section 2.3.4. For the initial finite element discretization, we discretize $\Omega = (0, \pi)^2$ with an unstructured Netgen mesh, with a maximum element diameter of 0.3 [27]. We take V_h to be the space of linear Lagrange elements satisfying the zero tangential trace boundary condition in (2.18). As before, to implement the adaptive mesh refinement technique in Algorithm 3, we also consider the enriched space V_h^+ , obtained by increasing the degree of V_h by one.

Being a two-dimensional problem, we expect the auxiliary eigensolves for the Maxwell operator to be more expensive than those required for the previous example. In an effort to speed up computations, we can further exploit the structure of the auxiliary eigenproblems.

Let $z = x + iy$, where $x, y \in \mathbb{R}$. For a self-adjoint operator, it can be shown that the adaptive mesh refinement technique in Algorithm 3 depends only on $x = \text{Re}(z)$.

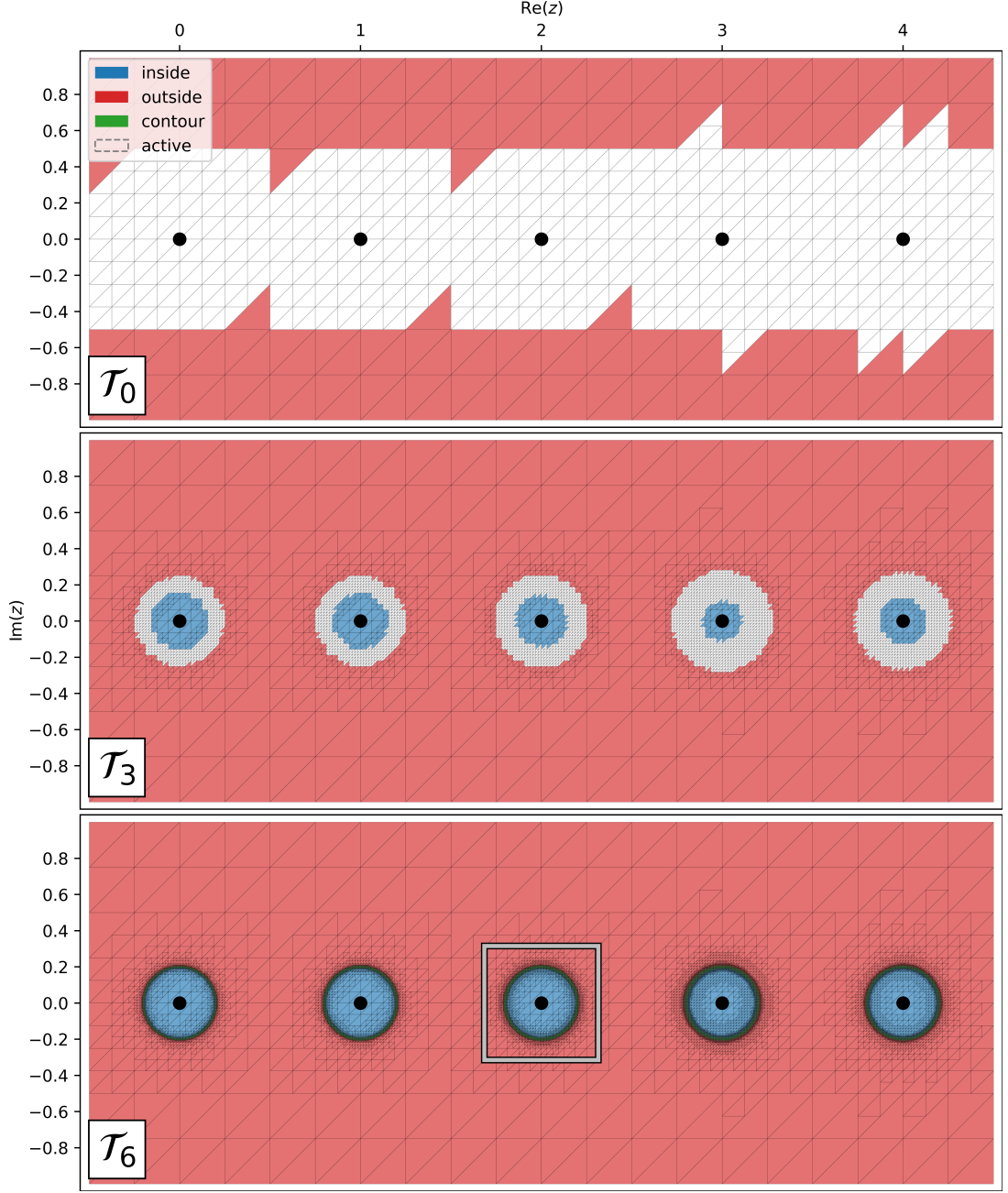
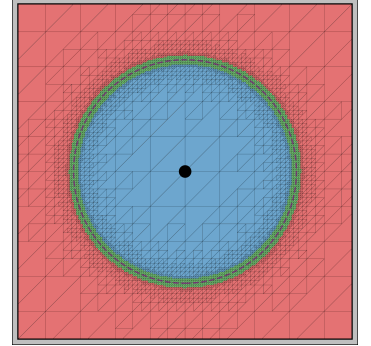


Figure 4.1: Outer-refinement levels $k = 0, 3, 6$ for Algorithm 3 applied to the $\epsilon = 0.2$ pseudospectrum of the Dirac-type operator (4.6). Black dots mark the exact eigenvalues (4.5). Red, blue, and green triangles are classified as outside, inside, and on the pseudospectral contour, respectively. White triangles are unclassified. A magnified view of the $\mathcal{T}_6$ result over $[1.7, 2.3] \times [-0.3, 0.3]$ is shown to the right, with the desired contour indicated by the dashed circle.



In particular, we have that

$$\Phi_h(z, A) = \sqrt{\mu_h(x) + y^2}, \quad (4.11)$$

where $\mu_h(x)$ is the smallest eigenvalue obtained from the adaptive mesh refinement technique applied to the folded operator at x . Thus, for all test points lying in the same column of the complex plane, only a single implementation of our adaptive mesh refinement technique is required. Further details can be found in Appendix C.

With this result in mind, to implement Algorithm 3 we iterate over each real barycentre z_K and employ the adaptive mesh refinement technique until the DWR error estimate $\eta(z_K)$ satisfies $|\eta(z_K)| \leq h_K^2$, or until the adapted meshes exceed a maximum of 100,000 degrees of freedom. We then use the result at z_K to compute $\Phi_h(z, A)$ for all other barycentres with the same real part as z_K .

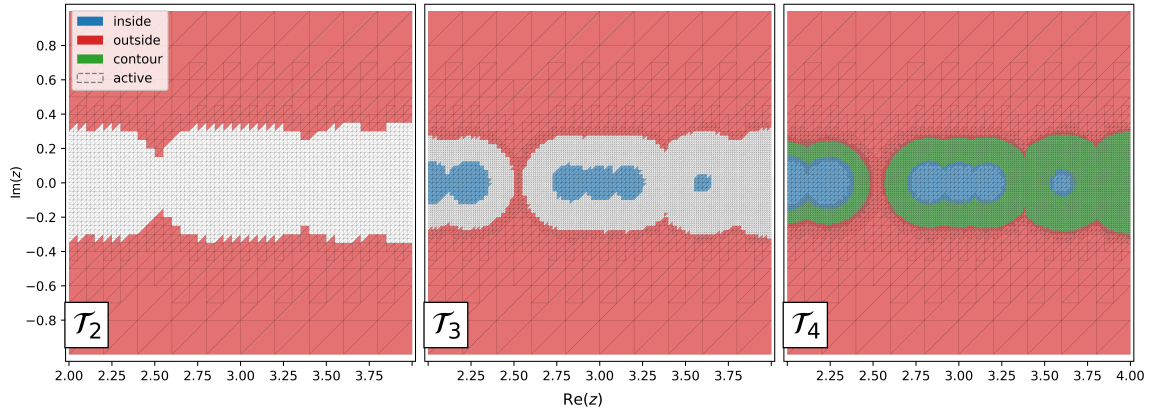


Figure 4.2: Outer-refinement levels $k = 2, 3, 4$, for Algorithm 3 applied to the $\epsilon = 0.2$ pseudospectrum of the Maxwell operator (2.66). The colour scheme is as in Figure 4.1.

The results of this numerical experiment are displayed in Figure 4.2. The algorithm performed five levels of outer refinement before the diameter of the remaining triangles met the outer tolerance and were taken to approximate the pseudospectral contour. Similar to the previous example, we observe a deterioration in the resolution of the pseudospectral contour as $\text{Re}(z)$ increases. Once again, we attribute this behaviour to the increasingly oscillatory nature of the eigenfunctions as $\text{Re}(z)$ increases.

As expected, the triangles in the complex plane are refined as we approach the desired ϵ -pseudospectral contour, thereby imposing stricter stopping criteria in the corresponding adaptive eigensolves. This further demonstrates our methodology's ability to focus the computational effort of Algorithm 2 near the pseudospectral contour of interest.

Chapter 5

Conclusions and Future Work

In this dissertation, we developed a methodology for accurate and efficient infinite-dimensional pseudospectral computations. To do so, we first explored Colbrook’s algorithm for infinite-dimensional pseudospectral computations. This approach characterizes the pseudospectrum of a closed and densely defined linear operator A using the smallest spectral values of the folded operators $(A - zI)^*(A - zI)$ and $(A - zI)(A - zI)^*$. In doing so, it avoids discretizing the operator A directly. Instead, it discretizes the auxiliary eigenproblems that characterize the smallest spectral values of the folded operators. This delayed discretization underpins the success of Colbrook’s algorithm, which avoids the common pitfalls of the standard discretize-then-solve approach.

After exploring the abstract formulation of Colbrook’s algorithm, we reformulated it for the setting in which the auxiliary eigenproblems are discretized with the finite element method. We then demonstrated the success of this approach by comparing it to the standard discretize-then-solve approach for the Maxwell eigenproblem.

These initial experiments revealed that, although the finite element realization of Colbrook’s algorithm avoids the pitfalls of the discretize-then-solve approach, its accuracy is limited by the resolution of the underlying auxiliary eigenproblems. To address this limitation, we devised an adaptive mesh refinement technique that uses DWR estimates for the auxiliary eigenvalue error to improve the accuracy of our finite element realization of Colbrook’s algorithm. This technique refines the finite element mesh in regions where the error estimate is large. It also uses the DWR estimates to correct Colbrook’s pseudospectral computations. By applying this approach to the Maxwell eigenproblem, we showed that these two uses of the DWR method can improve both the accuracy and efficiency of Colbrook’s approach.

Motivated by these results, we developed a methodology for coupling adaptive refinement of the auxiliary eigenproblems with adaptive refinement of the test points used in Colbrook’s algorithm. To do so, we first used the DWR eigenvalue error estimate to derive an error estimate for Colbrook’s pseudospectral computations. This estimate was used to formulate stopping criteria for our adaptive mesh refinement technique based on the desired accuracy of Colbrook’s pseudospectral computations.

We then developed a method for adaptively selecting the test points in Colbrook’s algorithm. We took these test points to be the barycentres of a triangulation of the complex plane. For each triangle, we formulated criteria for deciding whether the

triangle lies inside or outside the desired ϵ -pseudospectrum, or whether the triangle should be further refined. These classification criteria depend on both the diameter of the triangle and the DWR error estimate for Colbrook's pseudospectral computations. We therefore chose the stopping criterion for each adaptive eigensolve based on the diameter of the corresponding triangle. In this way, our methodology avoids solving the auxiliary eigenproblems to a greater accuracy than what is warranted by the current resolution of the complex plane.

The success of this coupling was demonstrated through its application to a one-dimensional Dirac-type operator and to the Maxwell operator. In both cases, our methodology focused the computational effort near the pseudospectral contour of interest.

In future work, we would like to apply our current methodology to non-normal operators with more complicated pseudospectra. One example is the advection-diffusion operator explored in [30]. To further increase the efficiency of our methodology, we would also like to investigate the use of multilevel methods for exploiting the hierarchy of meshes created by our adaptive mesh refinement technique.

We would also like to relax some of the underlying assumptions of our methodology so that it can be applied to a broader class of operators. In particular, we would like to extend the DWR method to estimate the error in computing general spectral values. We would also like to extend our finite element formulation of Colbrook's algorithm so that it can accommodate a wider range of operators.

Ultimately, the results of this dissertation demonstrate the potential of pairing adaptivity with Colbrook's algorithm for computing the pseudospectra of infinite-dimensional operators. We hope that the ideas presented here provide a foundation for further work toward accurate and efficient solvers for computing the pseudospectra of a broad class of operators.

Appendix A

The Spectra of the Shift Operators

In this appendix, we explore the spectra of the shift operators. To do so, we consider the sequences $x := \{x_n\}_{\mathbb{N}}$ and $y := \{y_n\}_{\mathbb{N}}$ in the $\ell^2(\mathbb{N})$ sequence space. This is a Hilbert space with norm satisfying $\|x\|_{\ell^2(\mathbb{N})}^2 = \sum_{n=1}^{\infty} |x_n|^2$ and with inner product $(x, y)_{\ell^2(\mathbb{N})} = \sum_{n=1}^{\infty} x_n \overline{y_n}$. Define the left and right shift operators on $\ell^2(\mathbb{N})$ as

$$L(x_1, x_2, \dots) = (x_2, x_3, \dots) \quad \text{and} \quad R(x_1, x_2, \dots) = (0, x_1, x_2, \dots), \quad (\text{A.1})$$

respectively.

Notice that $L^* = R$, and that their operator norms satisfy $\|R\| = \|L\| = 1$. When an operator A is bounded, we have that $\text{Sp}(A) \subset \{z \in \mathbb{C} \mid |z| \leq \|A\|\}$ [22]. Thus,

$$\text{Sp}(L), \text{Sp}(R) \subset \{z \in \mathbb{C} \mid |z| \leq 1\}.$$

The eigenvalues of L are found by solving $Lx = \lambda x$ for a $\lambda \in \mathbb{C}$ and a nonzero $x \in \ell^2(\mathbb{N})$. Thus, the eigenvectors are of the form

$$x = x_1(1, \lambda, \lambda^2, \dots), \quad (\text{A.2})$$

and must satisfy

$$|x_1|^2 \sum_{n=0}^{\infty} |\lambda|^{2n} < \infty. \quad (\text{A.3})$$

From this we find that

$$\Gamma(L) := \text{set of eigenvalues of } L = \{\lambda \in \mathbb{C} \mid |\lambda| < 1\} \subset \text{Sp}(L). \quad (\text{A.4})$$

Since the spectrum of a bounded operator is closed, the closed disk must also be in $\text{Sp}(L)$ [22]. Thus, $\text{Sp}(L) \supset \{z \in \mathbb{C} \mid |z| \leq 1\}$, $\text{Sp}(L) \subset \{z \in \mathbb{C} \mid |z| \leq 1\}$ and so

$$\text{Sp}(L) = \{z \in \mathbb{C} \mid |z| \leq 1\}.$$

The eigenvalues of R are found by solving $Rx = \lambda x$ for a $\lambda \in \mathbb{C}$ and a nonzero $x \in \ell^2(\mathbb{N})$. In doing so, we find that the eigenvectors must satisfy

$$(0, x_1, x_2, \dots) = \lambda(x_1, x_2, \dots). \quad (\text{A.5})$$

Equivalently,

$$\lambda x_1 = 0, \quad \lambda x_2 = x_1, \quad \dots \quad (\text{A.6})$$

The only solution is $x = 0$, which cannot be an eigenvector. Thus, R has no eigenvalues.

Recall that when an operator A is closed and densely defined, $\text{Sp}(A^*) = \{\bar{z} \mid z \in \text{Sp}(A)\}$ [9]. Here, L and R are bounded operators on all of $\ell^2(\mathbb{N})$ and $L^* = R$. Thus,

$$\text{Sp}(R) = \{\bar{z} \mid z \in \text{Sp}(L)\} = \{z \in \mathbb{C} \mid |z| \leq 1\}. \quad (\text{A.7})$$

That is, the spectrum of the right shift operator is the closed unit disc.

Appendix B

Choosing the Grid in Section 3.1.4

In this Appendix, we explore the effect of restricting $\text{Grid} \subset \mathbb{C}$ to complex numbers z satisfying $\text{Re}(z) \geq 2$. We consider this restriction when applying our adaptive mesh refinement technique of Section 3.1.3 to the auxiliary eigenproblems that arise when applying Algorithm 2 to the Maxwell operator (2.66).

Let A denote the self-adjoint Maxwell operator given by (2.66). Furthermore, let $F(z) := (A - zI)^*(A - zI) = (A - \bar{z}I)(A - zI)$ denote the folded operator for Colbrook's algorithm. Consider $u \in N(A)$, where $N(A)$ denotes the nullspace of A . Applying $F(z)$ to u yields

$$F(z)u = (A - \bar{z}I)(A - zI)u = (A - \bar{z}I)(-zu) = |z|^2u. \quad (\text{B.1})$$

Since $N(A)$ is infinite-dimensional, $|z|^2$ is an eigenvalue of the folded operator with infinite multiplicity.

More generally, if λ_j is a positive eigenvalue of A with corresponding eigenfunction u_j , then,

$$F(z)u_j = (A - \bar{z}I)(A - zI)u_j = |\lambda_j - z|^2u_j. \quad (\text{B.2})$$

For Colbrook's algorithm, we want to compute the bottom of the spectrum of $F(z)$. To avoid the eigenvalue $|z|^2$ of infinite multiplicity, we want to ensure that there is at least one eigenvalue of the folded operator that is smaller than $|z|^2$. To do so, we want to ensure that there is a positive eigenvalue λ_j of A that satisfies

$$|\lambda_j - z|^2 < |z|^2. \quad (\text{B.3})$$

Setting $z = x + iy$, we can write this last inequality as

$$|(\lambda_j - x) - iy|^2 < x^2 + y^2. \quad (\text{B.4})$$

The left-hand side can be expanded, and the y^2 can be cancelled, to give

$$(\lambda_j - x)^2 < x^2, \quad (\text{B.5})$$

or

$$\lambda_j^2 - 2\lambda_j x < 0. \quad (\text{B.6})$$

Since $\lambda_j > 0$, the desired condition is

$$x > \lambda_j/2. \quad (\text{B.7})$$

Thus, if there exists a nonzero eigenvalue of A such that $\operatorname{Re}(z) > \lambda_j/2$, then $|z|^2$ will not be the smallest eigenvalue of the folded operator. The smallest nonzero eigenvalue of A is $\lambda = 1$. So, as long as we can guarantee that for every test point z , $\operatorname{Re}(z) > 1/2$, then $|z|^2$ will not be at the bottom of the spectrum of $F(z)$. Following this line of thought, for the numerical experiments of this report, we considered test points z satisfying $\operatorname{Re}(z) \geq 2$. In this way, we avoid computing the infinite-multiplicity eigenvalue $|z|^2$ at the bottom of the spectrum.

Appendix C

Self-Adjoint Speed Ups

In this appendix, we derive the result used to speed up our implementation of Algorithm 3 in Section 4.2.

Let A be a self-adjoint, closed, and densely defined operator on the Hilbert space $\mathcal{H}$. For Colbrook's algorithm, we consider the folded operator $F(z) := (A - zI)^*(A - zI)$. Since $A^* = A$, we have $F(z) = (A - \bar{z}I)(A - zI)$. Plugging in $z = x + iy$ and expanding, we obtain

$$F(z) = (A - xI + iyI)(A - xI - iyI) \quad (\text{C.1})$$

$$= (A - xI)^2 + y^2I. \quad (\text{C.2})$$

Next we consider the quadratic form

$$Q_{z,A}(u, v) := ((A - zI)u, (A - zI)v)_{\mathcal{H}}, \quad (\text{C.3})$$

for all $u, v \in D(A)$. Substituting in $z = x + iy$, we have

$$Q_{z,A}(u, v) = ((A - xI)u - iyu, (A - xI)v - iyv)_{\mathcal{H}} \quad (\text{C.4})$$

$$= ((A - xI)u, (A - xI)v)_{\mathcal{H}} + iy((A - xI)u, v)_{\mathcal{H}} - iy(u, (A - xI)v)_{\mathcal{H}} + y^2(u, v)_H \quad (\text{C.5})$$

$$= ((A - xI)u, (A - xI)v)_{\mathcal{H}} + y^2(u, v)_H \quad (\text{C.6})$$

$$= Q_{x,A}(u, v) + y^2(u, v)_{\mathcal{H}}. \quad (\text{C.7})$$

Taking $v = u$, we have that

$$Q_{z,A}(u, u) = Q_{x,A}(u, u) + y^2\|u\|_{\mathcal{H}}^2. \quad (\text{C.8})$$

Then, taking the infimum over unit vectors, we can write (2.25) as

$$\gamma(z, A)^2 = \gamma(x, A)^2 + y^2. \quad (\text{C.9})$$

Considering an appropriate finite element space V_h , we have that

$$Q_{z,A}(u_h, v_h) = Q_{x,A}(u_h, v_h) + y^2m(u_h, v_h). \quad (\text{C.10})$$

Thus, to seek $\lambda_h(z)$ and nonzero $u_h(z) \in V_h$ to satisfy

$$Q_{z,A}(u_h(z), v_h) = \lambda_h(z)m(u_h(z), v_h), \quad (\text{C.11})$$

for all $v_h \in V_h$, we can instead seek $\lambda_h(x)$ and nonzero $u_h(x) \in V_h$ to satisfy

$$Q_{x,A}(u_h(x), v_h) = (\lambda_h(x) - y^2)m(u_h(x), v_h) = \lambda_h(x)m(u_h(x), v_h), \quad (\text{C.12})$$

for all $v_h \in V_h$. In other words, for each distinct x , we can solve $Q_{x,A}(u_h(x), v_h) = \lambda_h(x)m(u_h(x), v_h)$ and find $\lambda_h(x)$ by adding y^2 . The eigenfunctions carry over as well. That is, $E_h(z) = E_h(x)$, where $E_h(\cdot)$ denotes the eigenspace for $\lambda_h(\cdot)$.

With this, we can conclude that for z values that share the same real part, Colbrook's bound on the distance to the spectrum depends only on a shift by the square of the imaginary part:

$$\Phi_h(z, A) = \sqrt{\mu_h(z)} = \sqrt{\mu_h(x) + y^2}, \quad (\text{C.13})$$

where $\mu_h(z)$ and $\mu_h(x)$ denote the smallest eigenvalues of (C.11) and (C.12), respectively.

This result carries over to the DWR error estimate (2.78) as well. In particular, notice that the primal residual (2.77) can be written as

$$\rho(u_h, \lambda_h(z))(v) := Q_{z,A}(u_h, v) - \lambda_h(z)m(u_h, v) \quad (\text{C.14})$$

$$= Q_{x,A}(u_h, v) + y^2m(u_h, v) - \lambda_h(z)m(u_h, v) \quad (\text{C.15})$$

$$= Q_{x,A}(u_h, v) - (\lambda_h(z) - y^2)m(u_h, v) \quad (\text{C.16})$$

$$= Q_{x,A}(u_h, v) - \lambda_h(x)m(u_h, v) \quad (\text{C.17})$$

$$= \rho(u_h, \lambda_h(x))(v). \quad (\text{C.18})$$

Since the primal residual depends only on x , as well as both the primal and enriched eigenspaces, the DWR error estimate (2.78) depends only on x . Thus,

$$\lambda(z) - \lambda_h(z) \approx \frac{\rho(u_h(z), \lambda_h(z))(u_h^+(z) - u_h(z))}{1 - 1/2\|u_h^+ - u_h\|_m^2} \quad (\text{C.19})$$

$$= \frac{\rho(u_h(x), \lambda_h(x))(u_h^+(x) - u_h(x))}{1 - 1/2\|u_h^+(x) - u_h(x)\|_m^2}. \quad (\text{C.20})$$

These results allow us to drastically speed up our implementation of Algorithm 3. For each distinct x , we perform one expensive adaptive eigensolve at one barycentre z_K that satisfies $\text{Re}(z_K) = x$. We then use (C.13) to compute $\Phi_h(z_K, A)$ for every other barycentre with the same real part. The DWR error estimate computed at the representative z_K is also carried over to every other barycentre with the same real part.

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